

An Introduction to the Smith Ecophysiology Lab: Physiology for Understanding Larger Scale Processes

Nick Smith

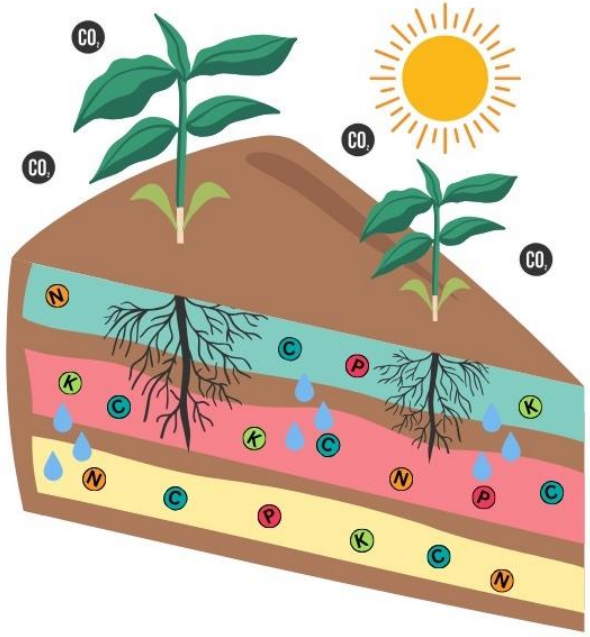
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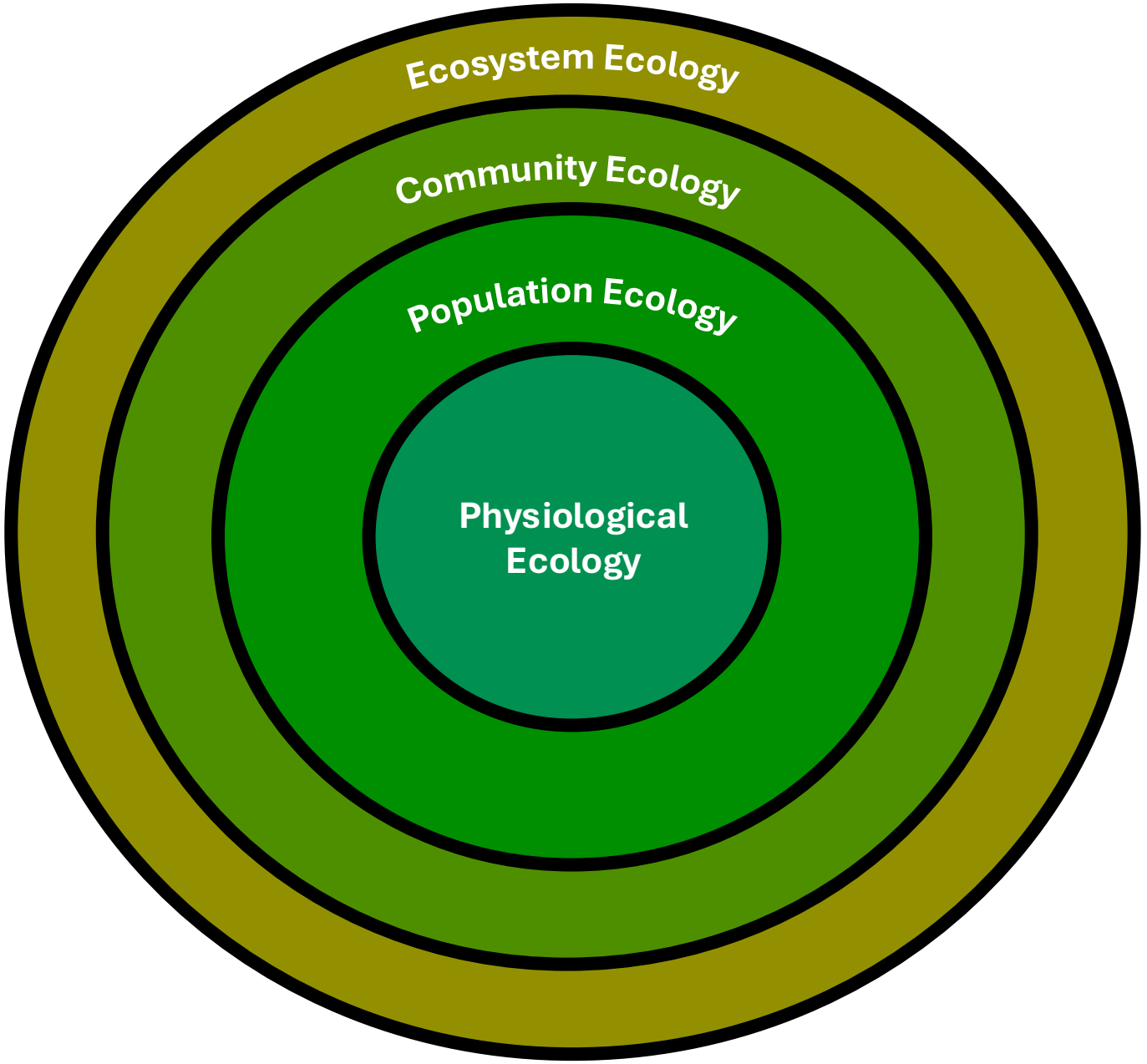
PhUnKY Lab

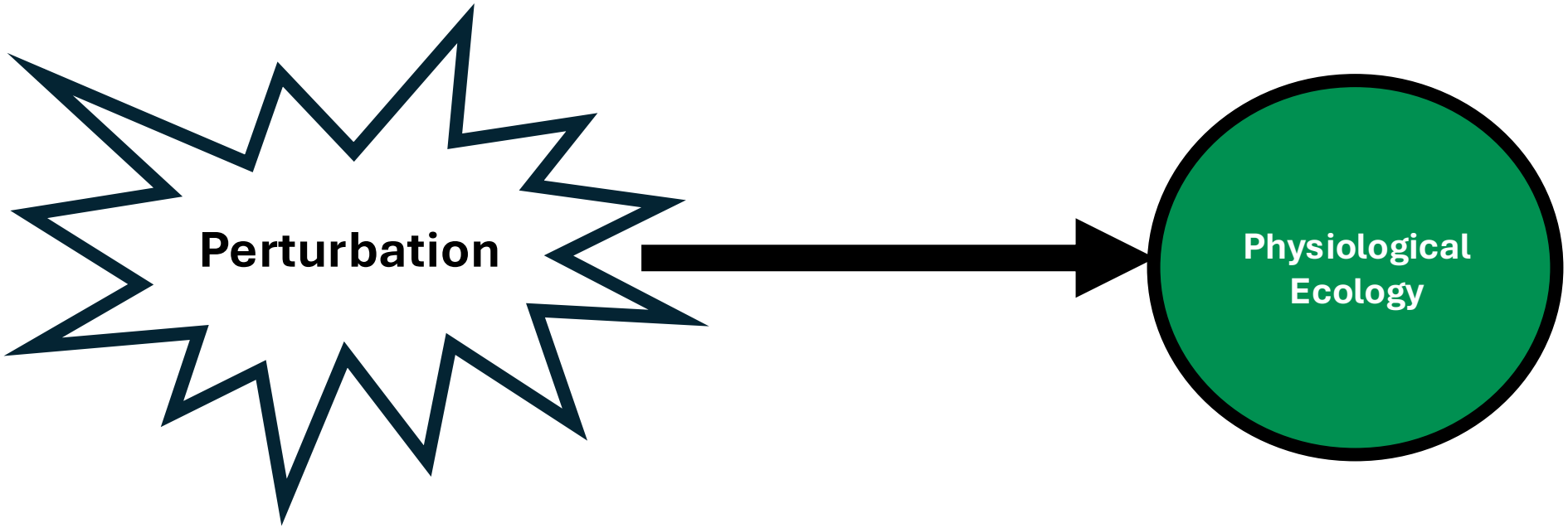
Physiology for Understanding and Knowledge in EcologY



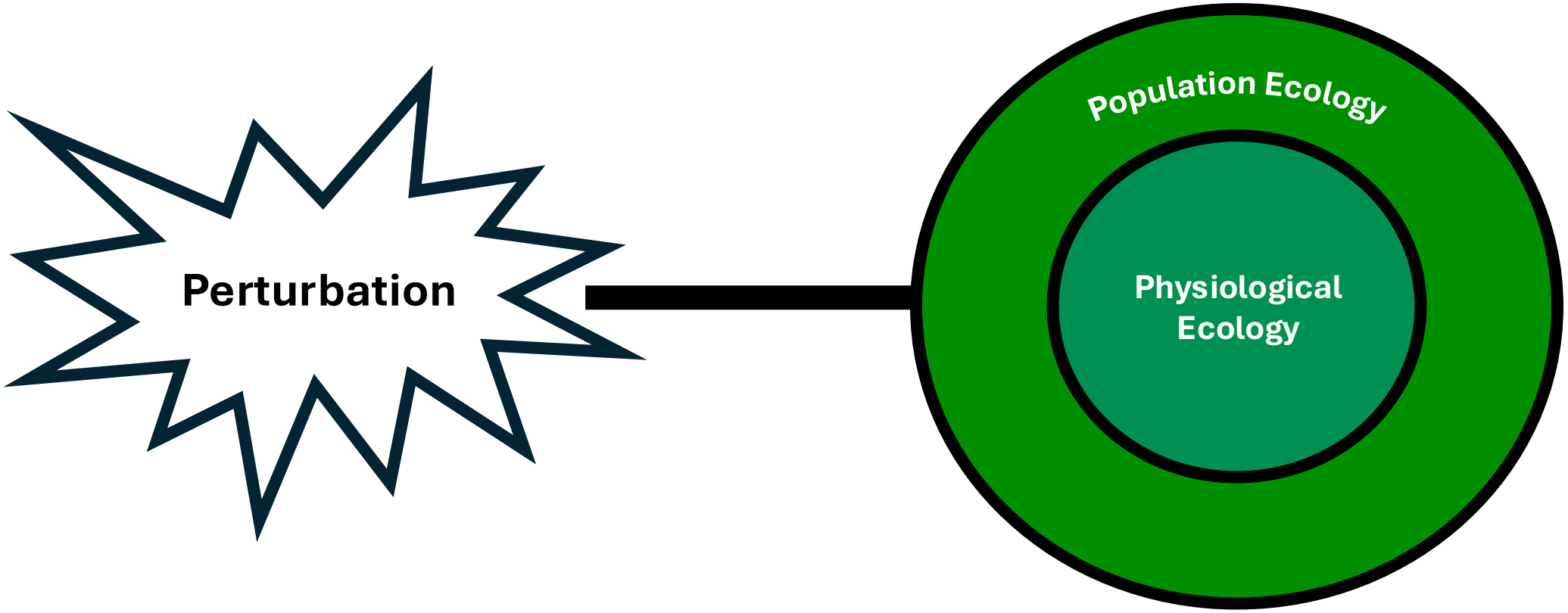
smithecophyslab.com

Ecological scales

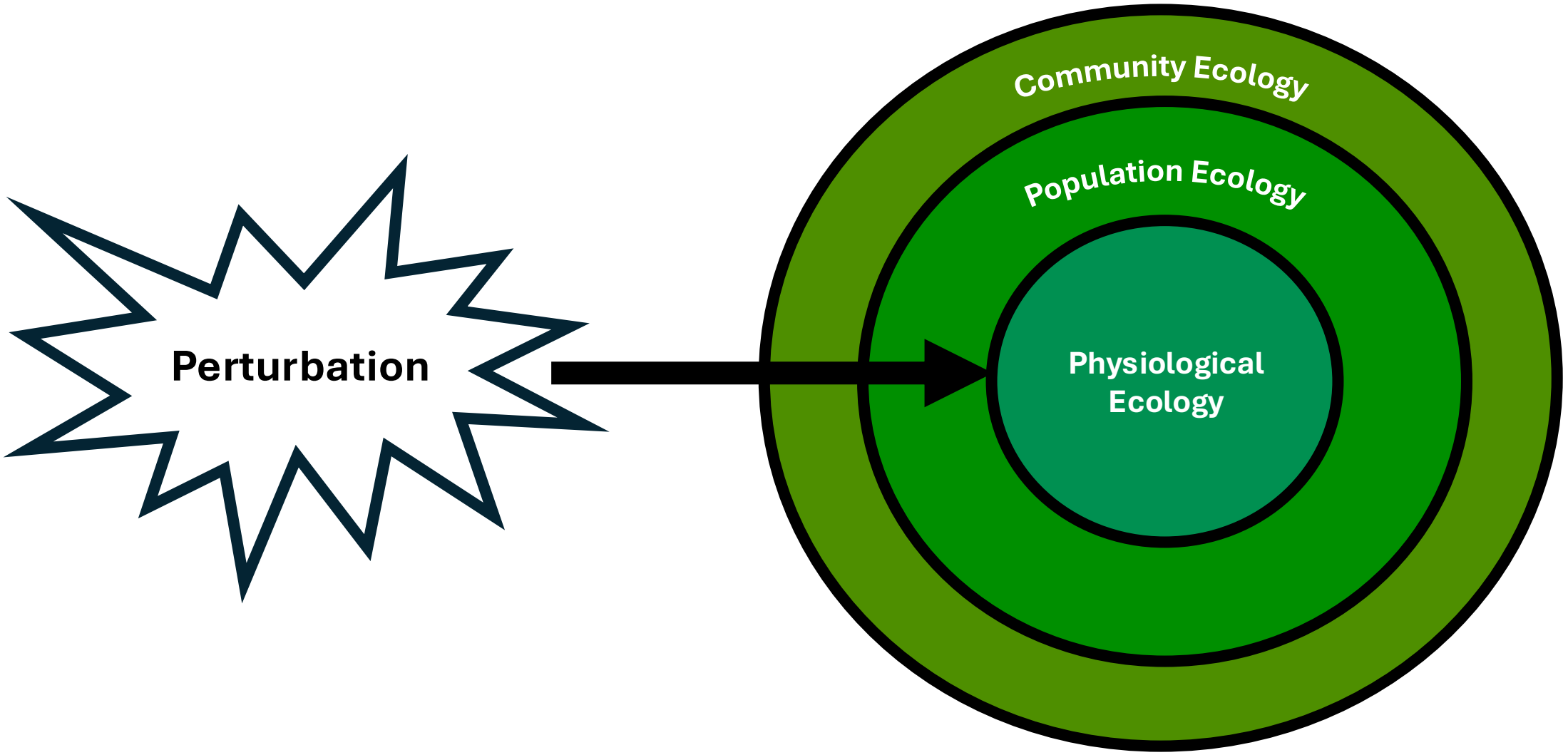




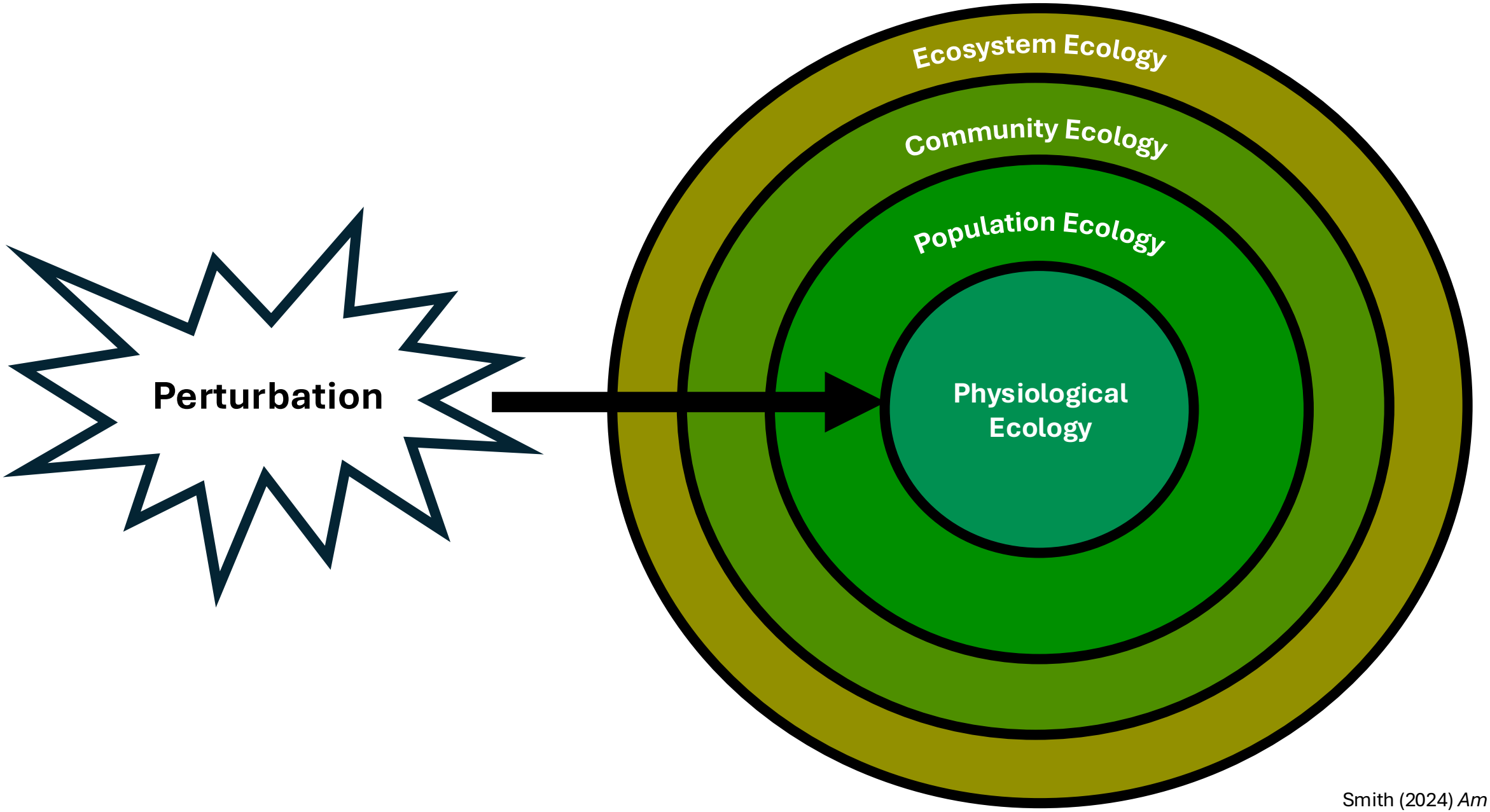
Smith (2024) *Am J Botany*



Smith (2024) *Am J Botany*



Smith (2024) *Am J Botany*



Smith (2024) *Am J Botany*

Structure of the talk

- **Short vignettes** illustrating connections between physiological and larger scale ecological processes under different **perturbations**
- Goal is to **highlight different questions and approaches**
- Would be happy to discuss any in more detail after the talk

Approach icons

Manipulation experiments





Field observations

Theory development





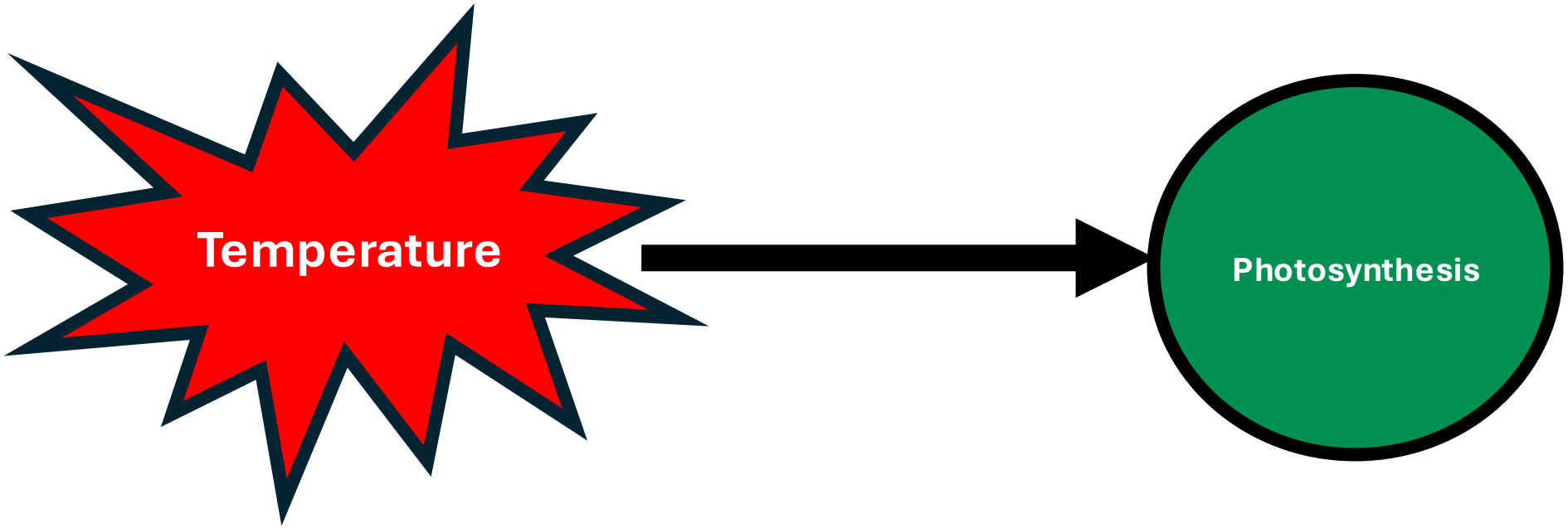
Process-based modeling

Perturbation vignettes

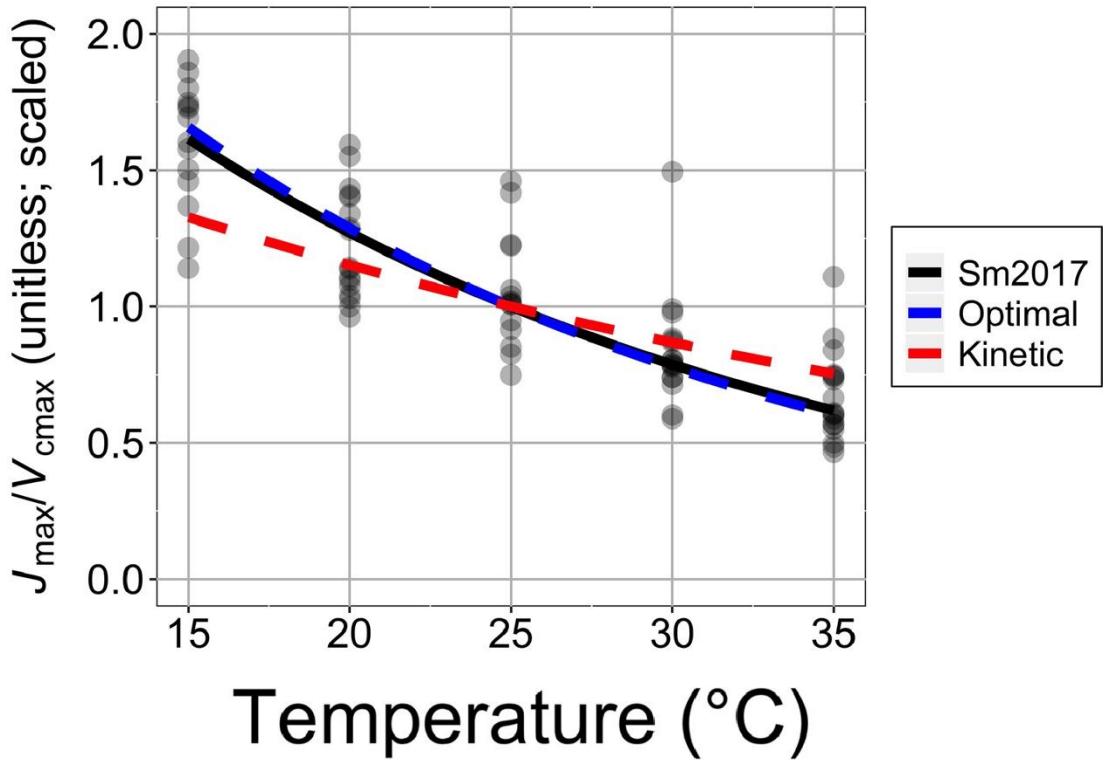
- **Temperature → global carbon and energy fluxes**
- **CO_2 → plant-nutrient dynamics**
- **Soil fertility → population dynamics**
- **Environmental heterogeneity → community dynamics**
- **Invasion → mycorrhizal symbioses**

Perturbation vignettes

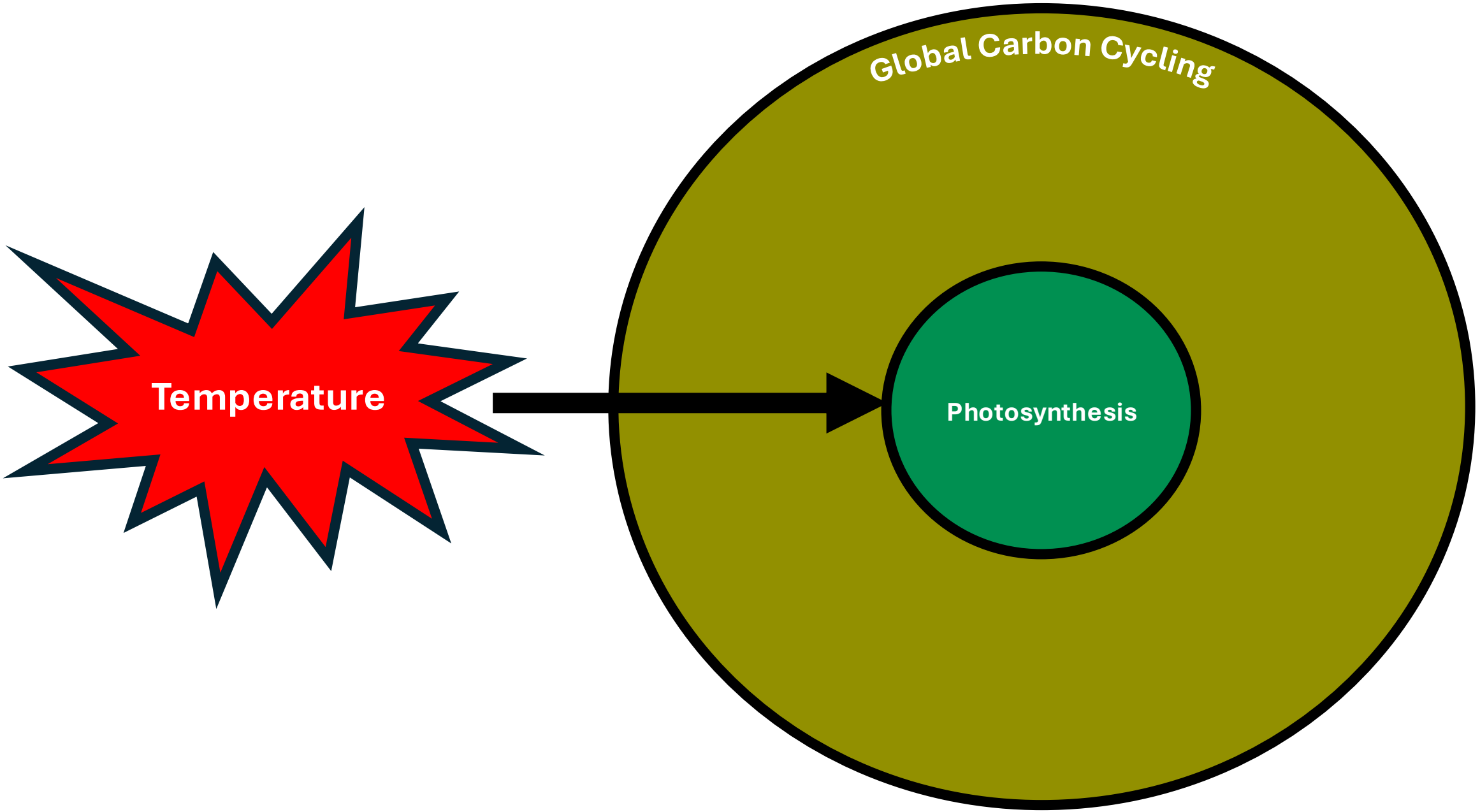
- **Temperature** → global carbon and energy fluxes
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Temperature
Photosynthesis

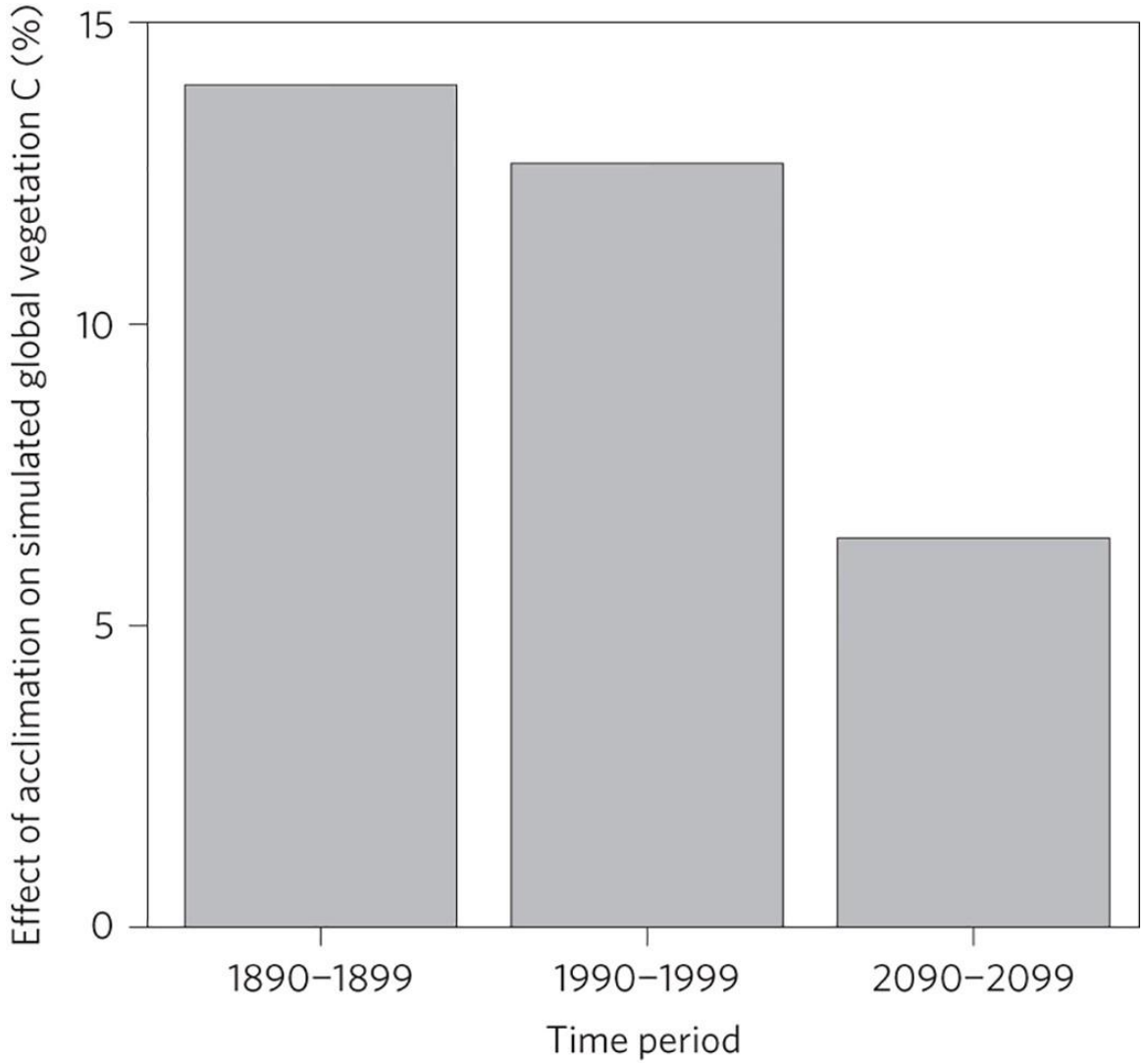
Photosynthesis shows strong and predictable acclimation responses to temperature



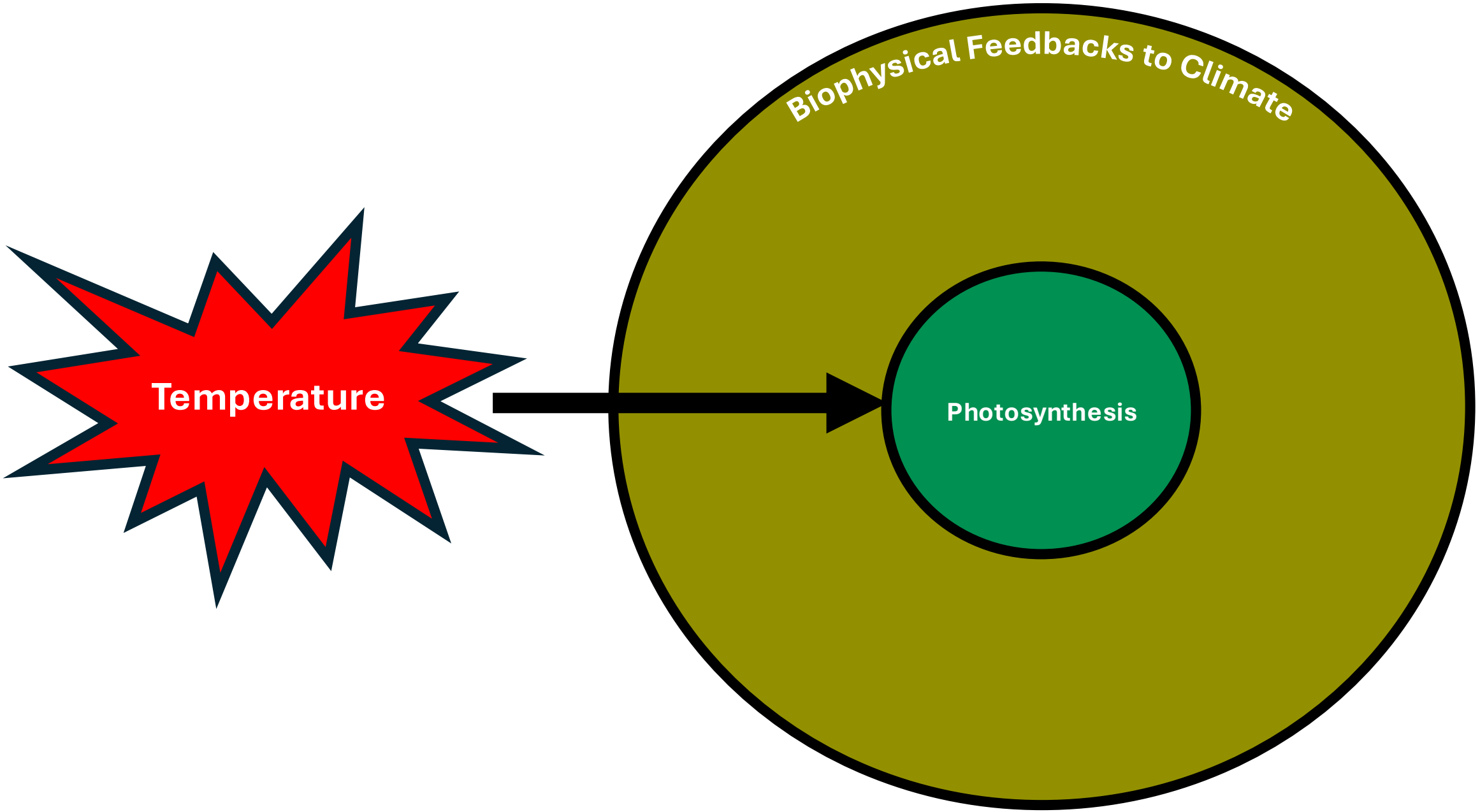
Smith and Keenan (2020) *Glob Change Bio*


Global Carbon Cycling
Temperature
Photosynthesis

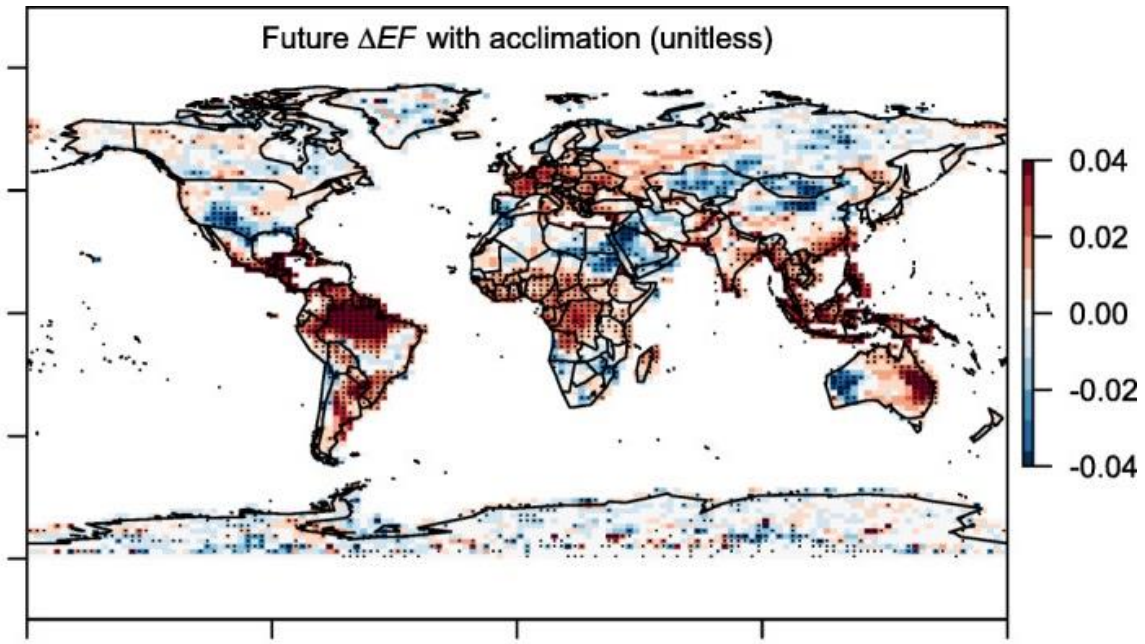
Photosynthetic temperature acclimation increases simulated global vegetation carbon

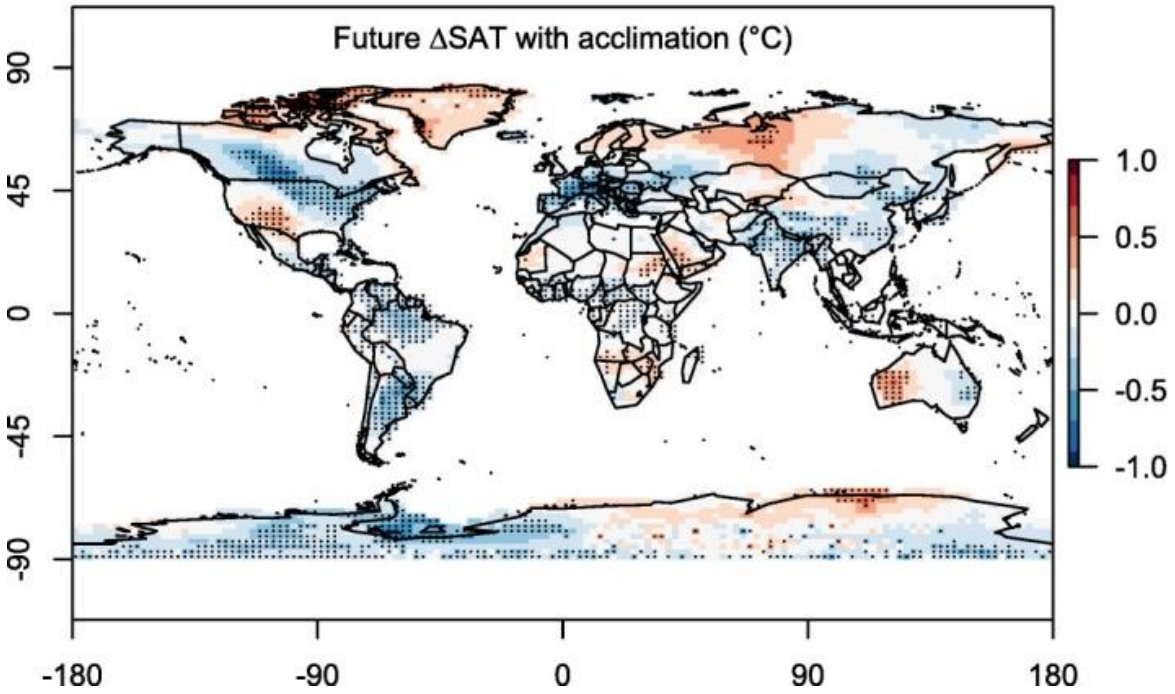


Smith et al. (2016) *Nat Clim Change*


Biophysical Feedbacks to Climate
Temperature
Photosynthesis

Photosynthetic temperature acclimation changes transpiration and air temperature

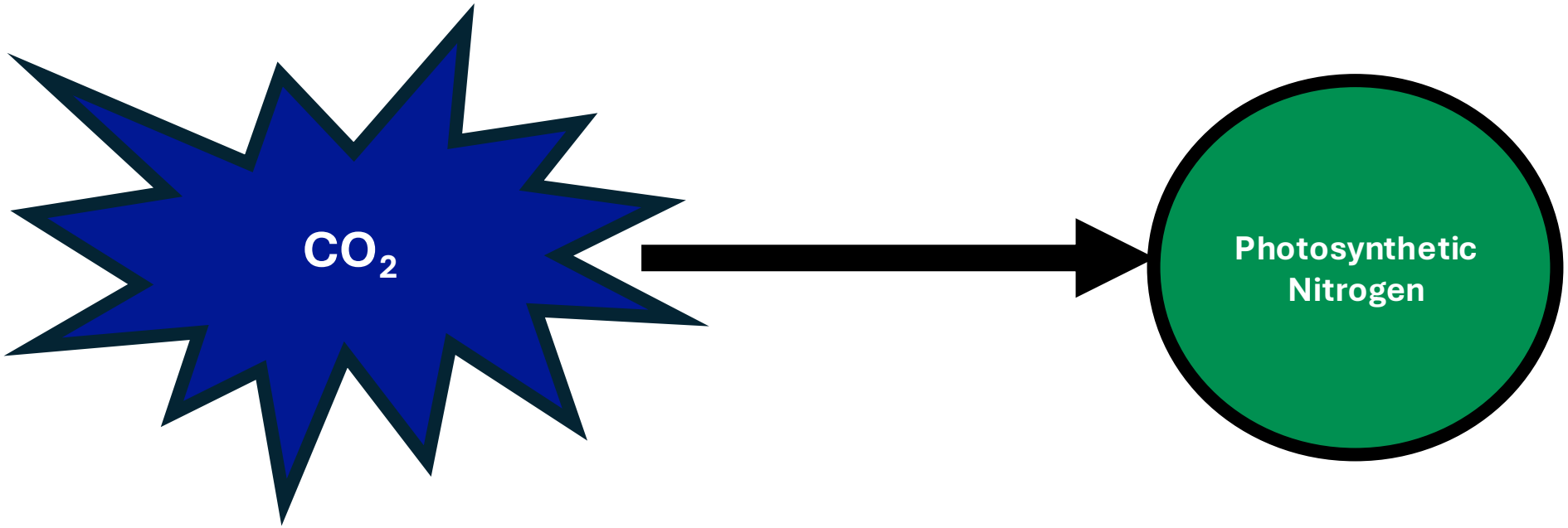




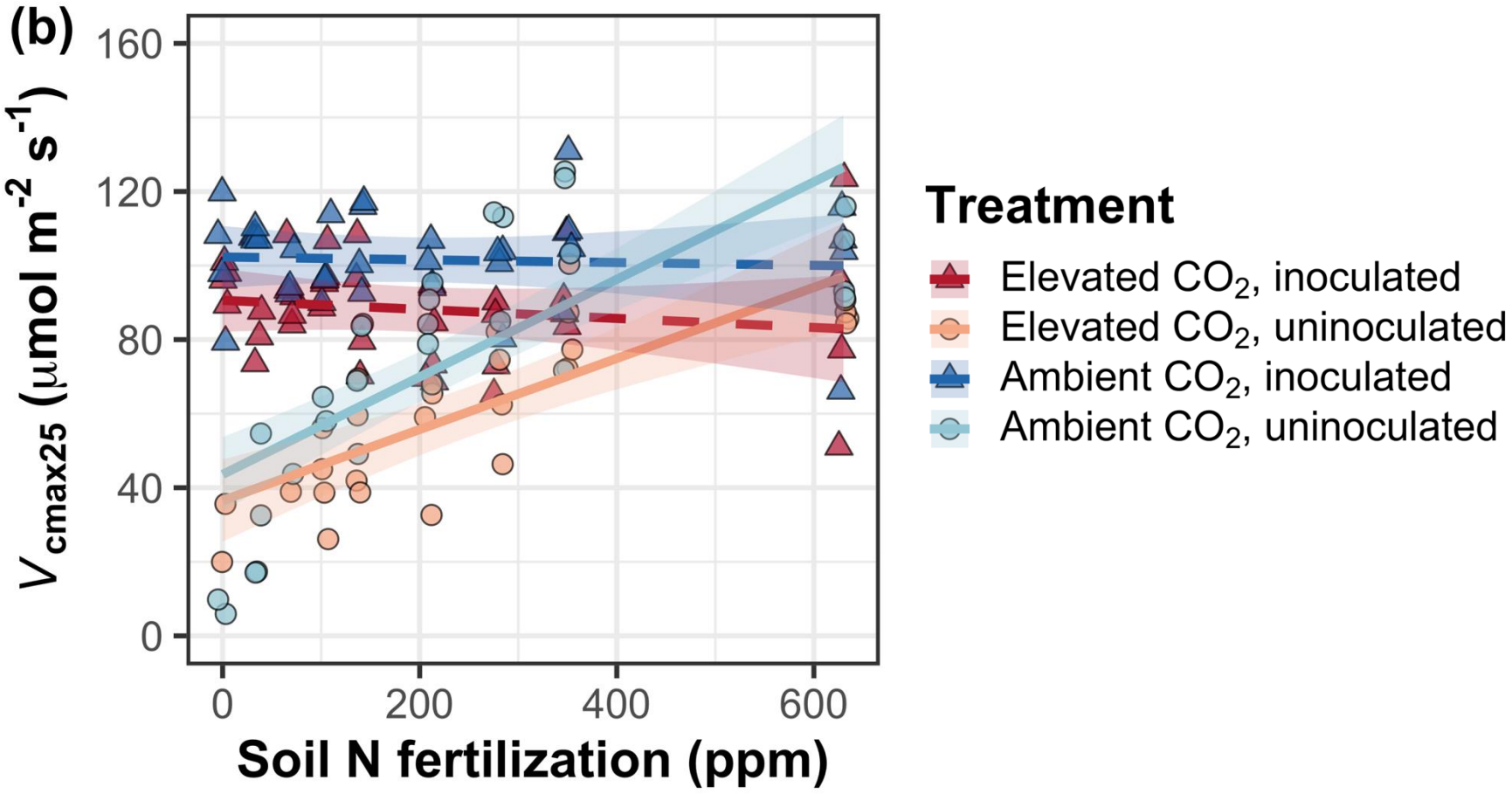
Smith et al. (2017) *JAMES*

Perturbation vignettes

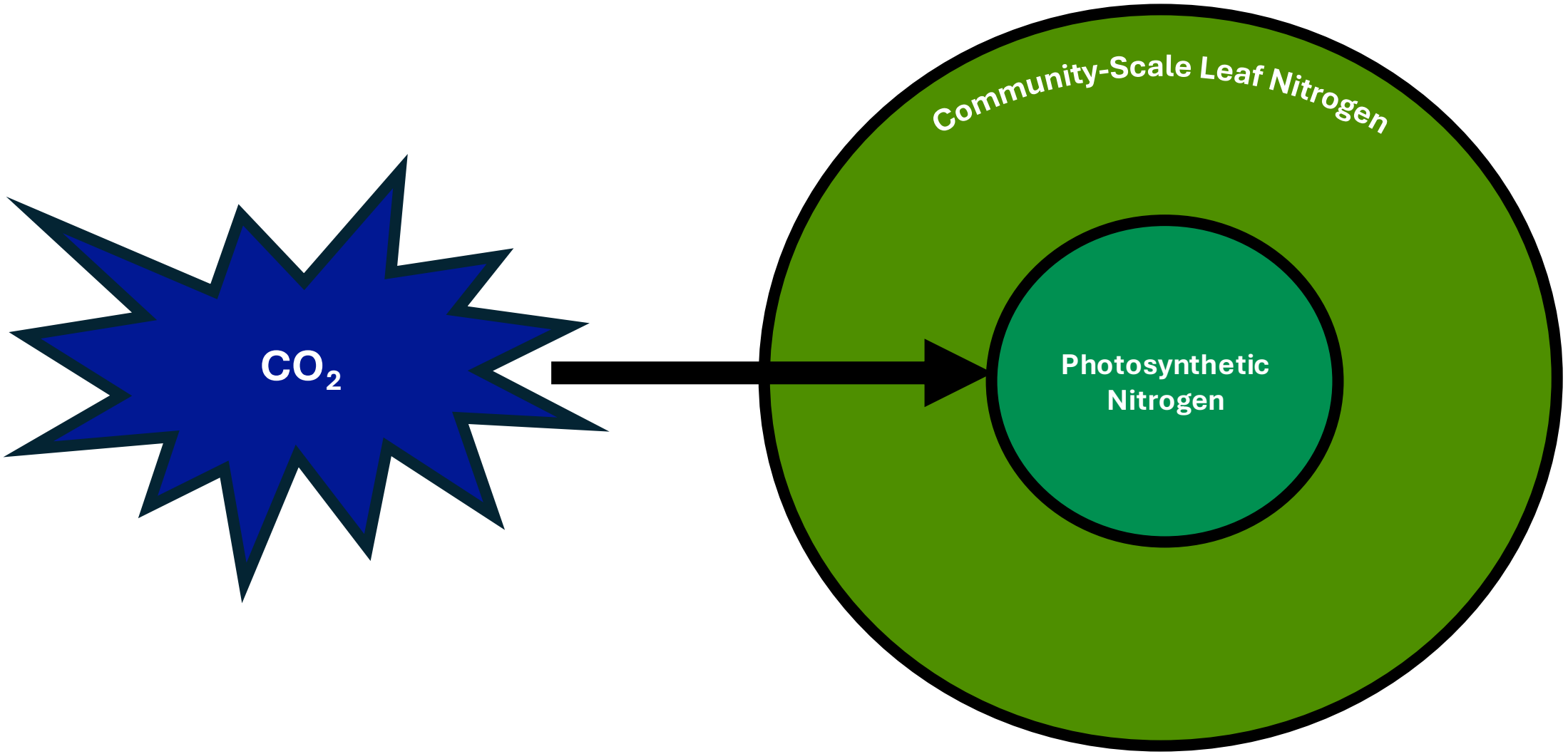
- Temperature → global carbon and energy fluxes
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CO_2
Photosynthetic
Nitrogen

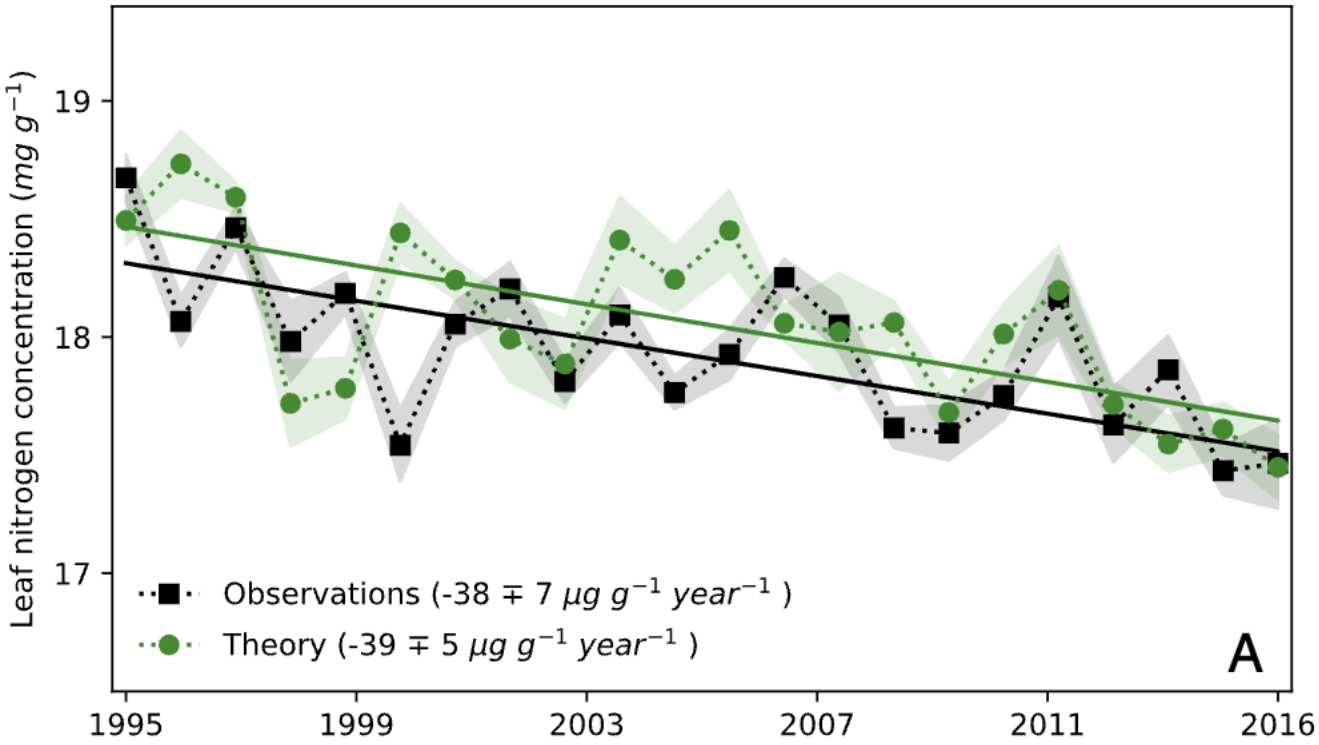
Elevated CO_2 reduces leaf nitrogen demand regardless of nitrogen availability



Perkowski, Ezekannagha, Smith (2025) *J Exp Bot*


CO_2
Community-Scale Leaf Nitrogen
Photosynthetic Nitrogen

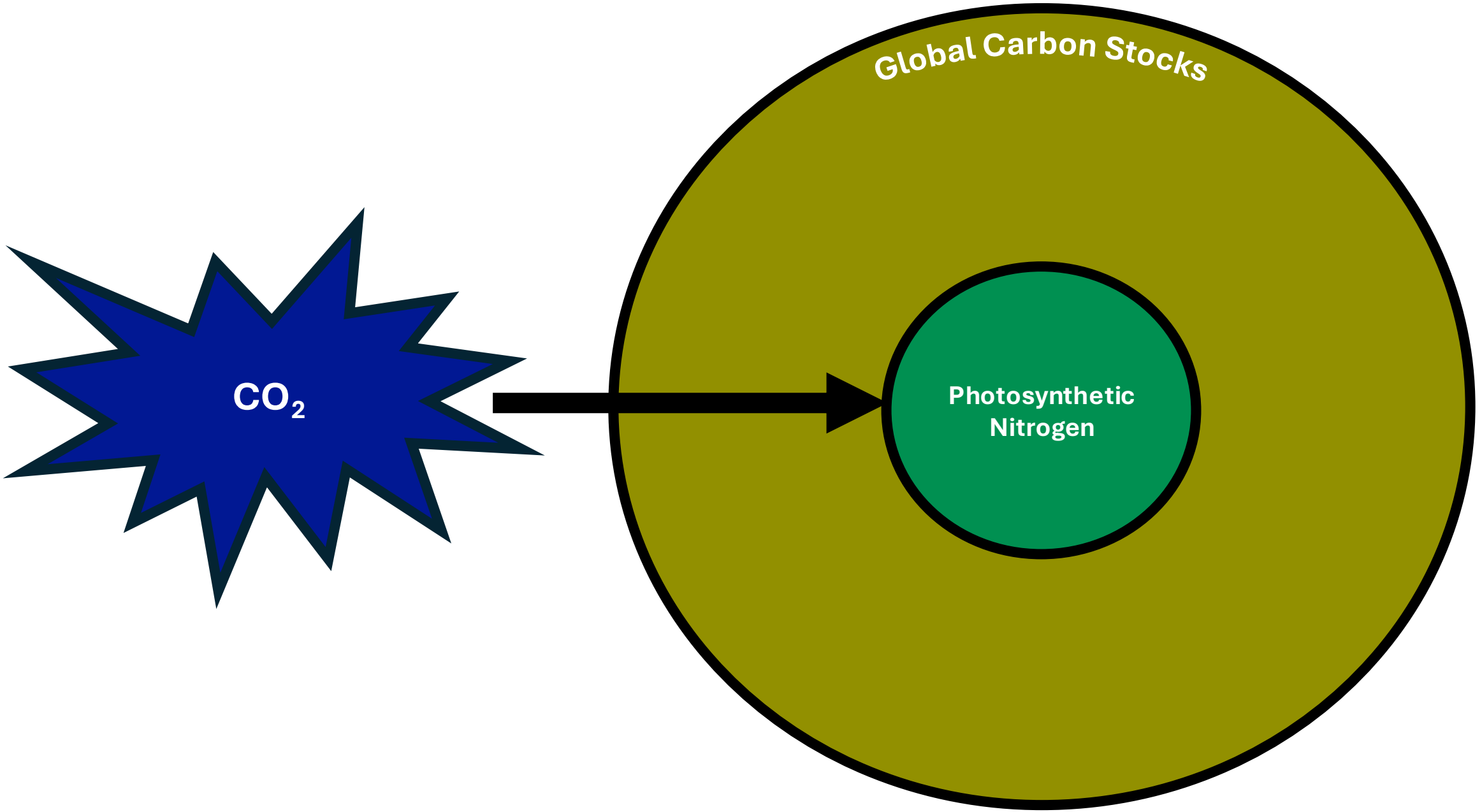
This is predictable over continental gradients



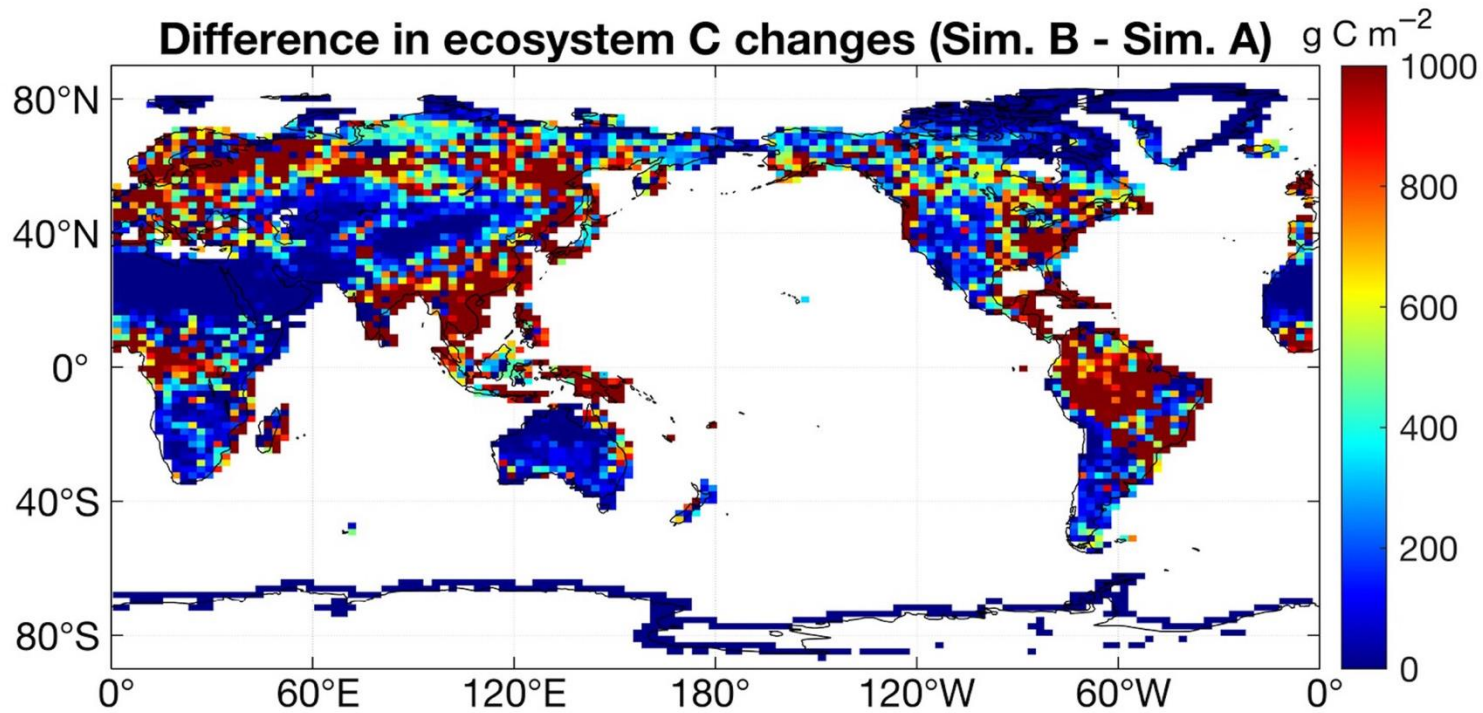




Bassiouni, Smith, et al. (2025) *PNAS*

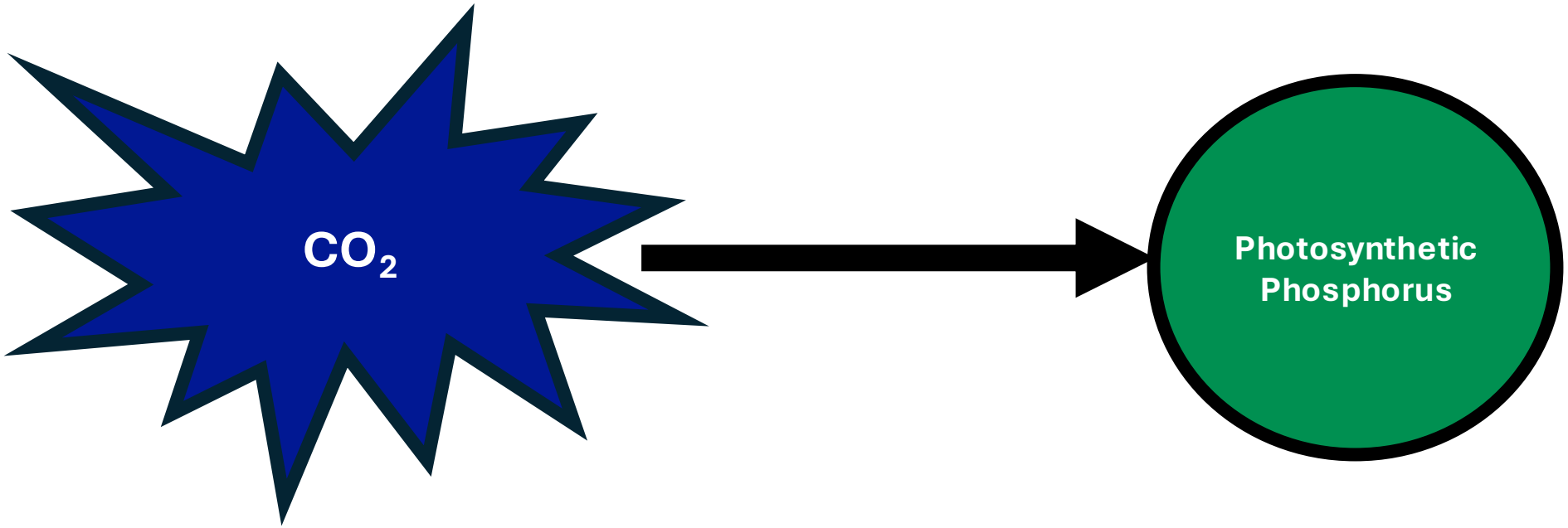

Global Carbon Stocks
CO_2
Photosynthetic Nitrogen

Accounting for this effect in models increases simulated future global carbon stocks

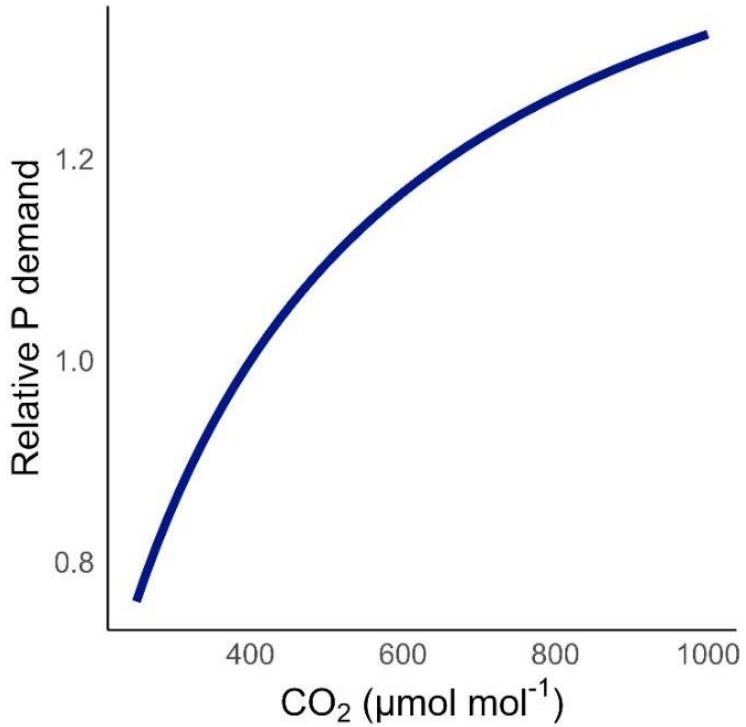


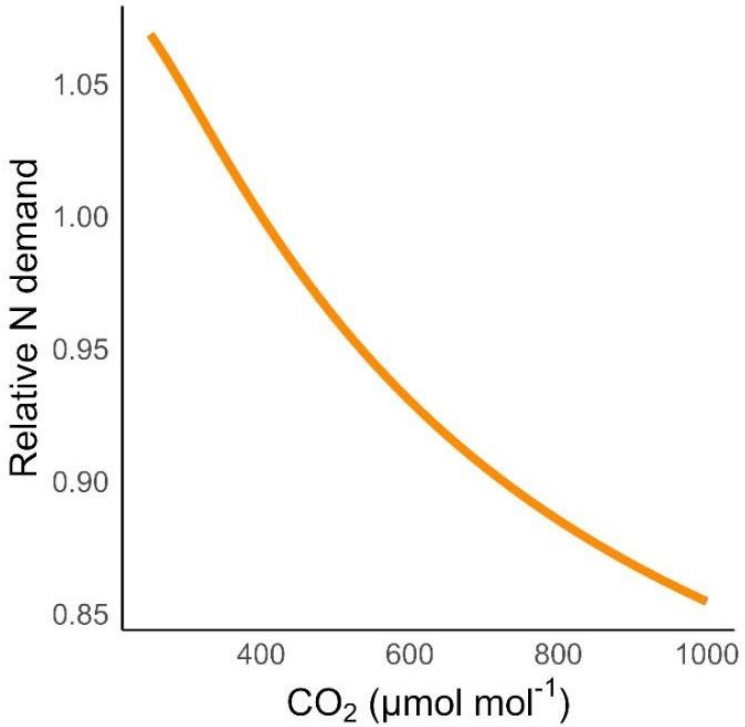


Smith et al. (2024) *Glob Change Biol*


CO_2
Photosynthetic
Phosphorus

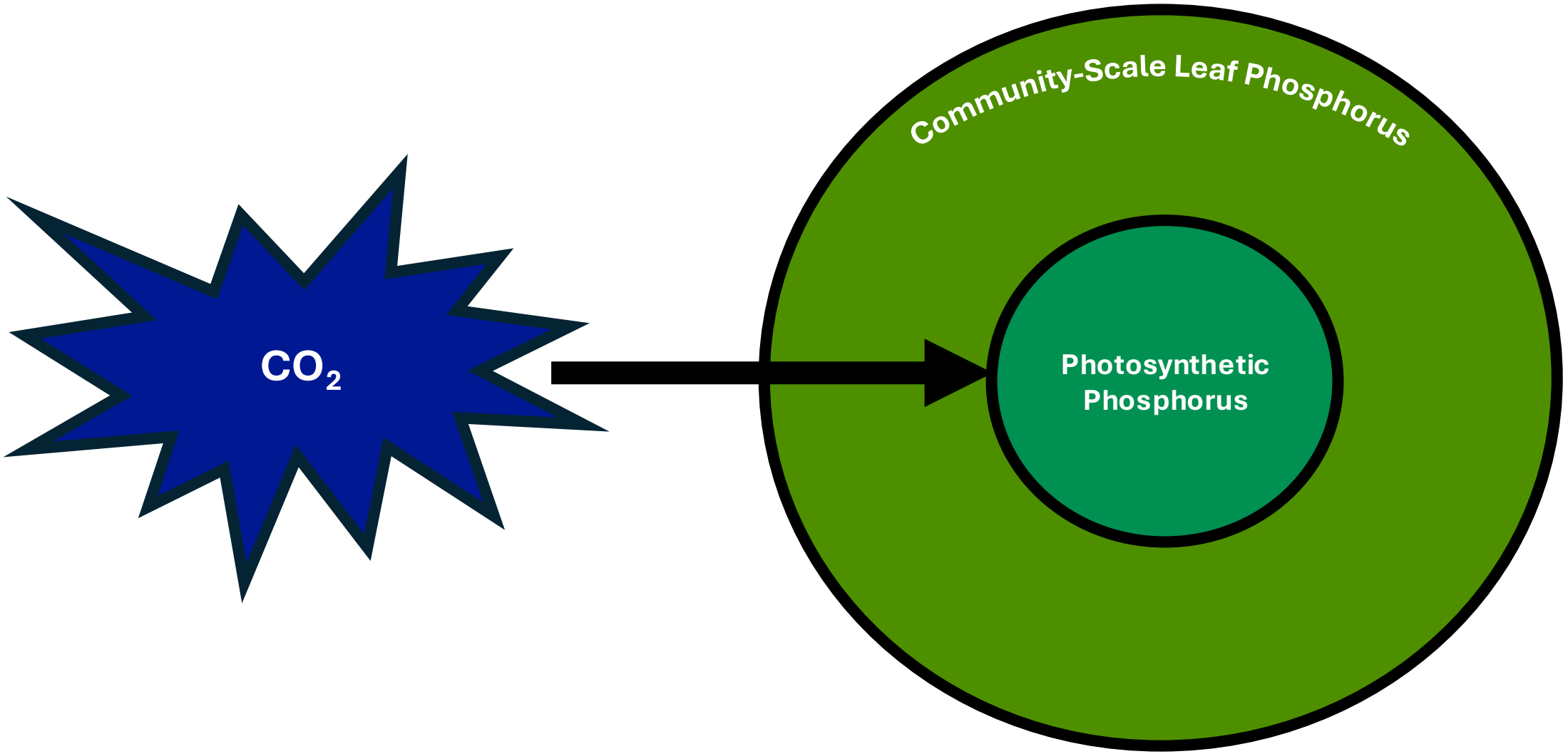
Phosphorus demand response differs from nitrogen demand response to CO_2



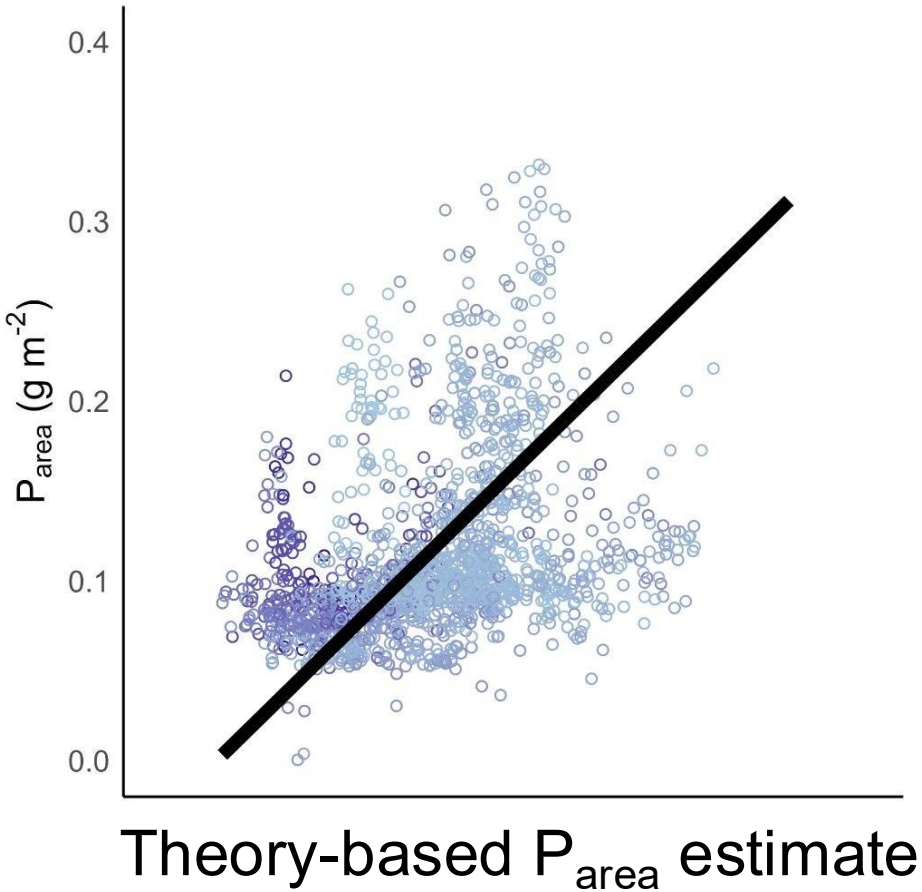




Smith et al. (In Prep)


CO_2
Community-Scale Leaf Phosphorus
Photosynthetic Phosphorus

Theory can predict community-scale leaf phosphorus



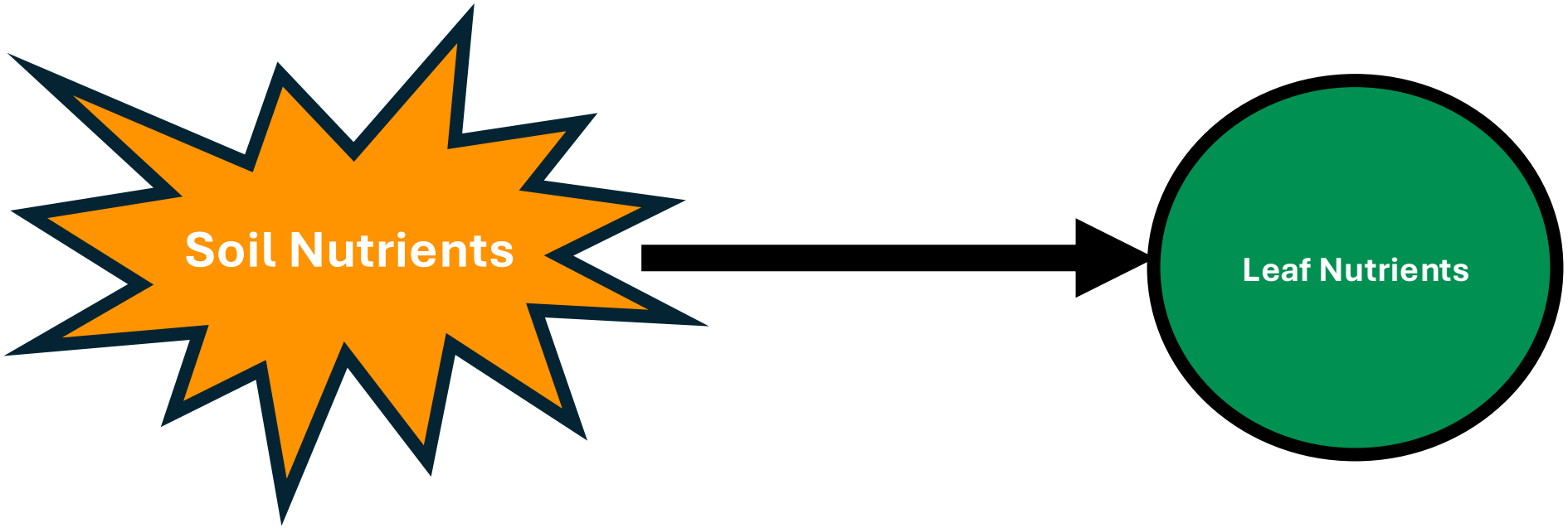




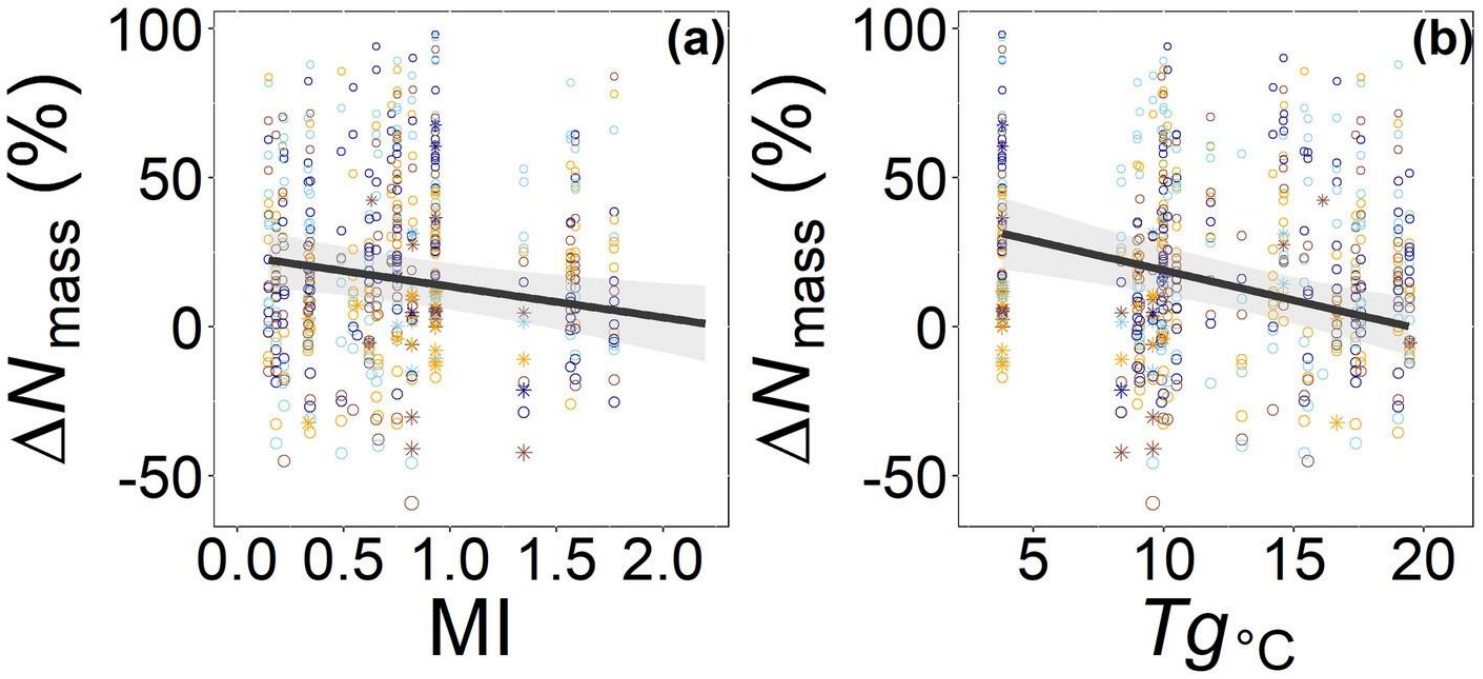
Smith et al. (In Prep)

Perturbation vignettes

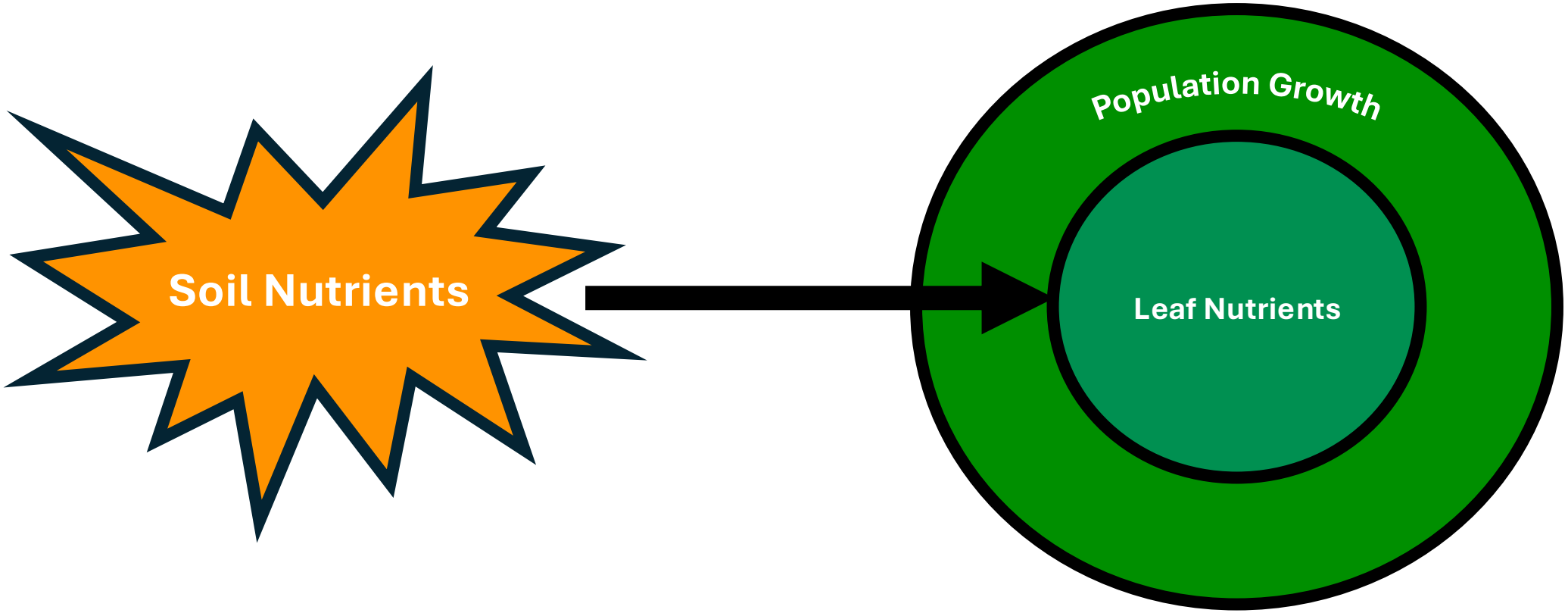
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Soil Nutrients
Leaf Nutrients

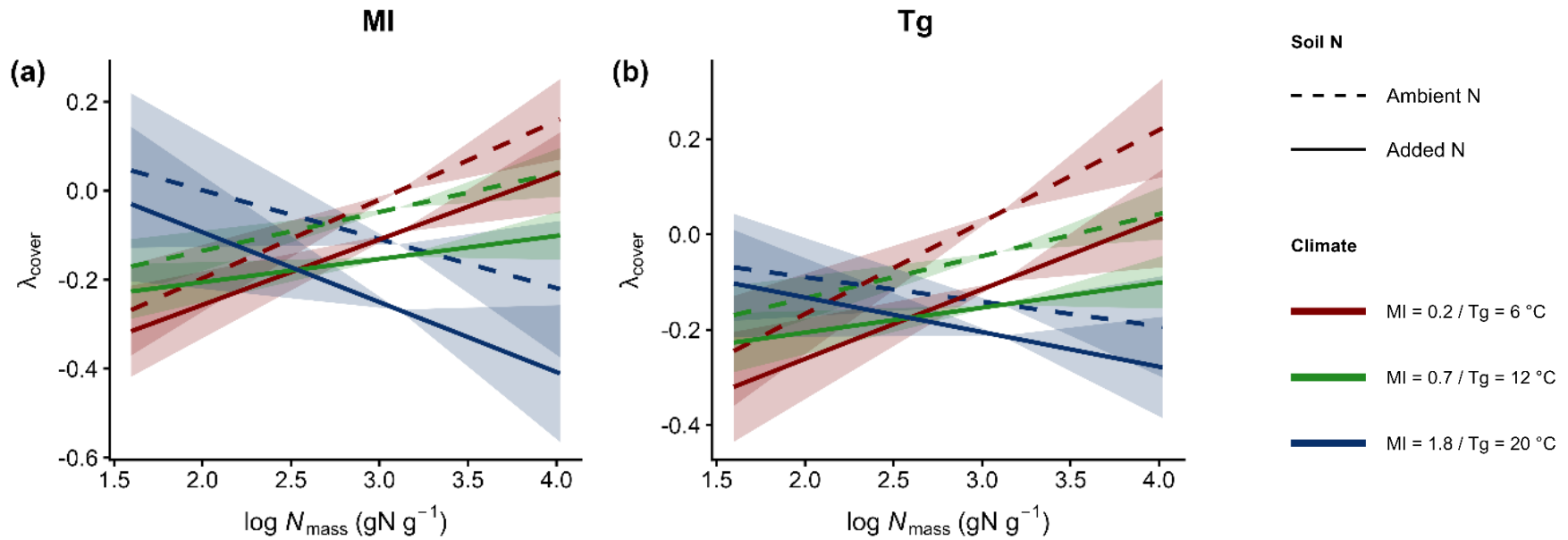
Soil nutrients enhance leaf nutrients where photosynthetic demand is greatest (dry/cold)



Cheaib…and Smith (2025) *Ecol Letters*


Soil Nutrients
Population Growth
Leaf Nutrients

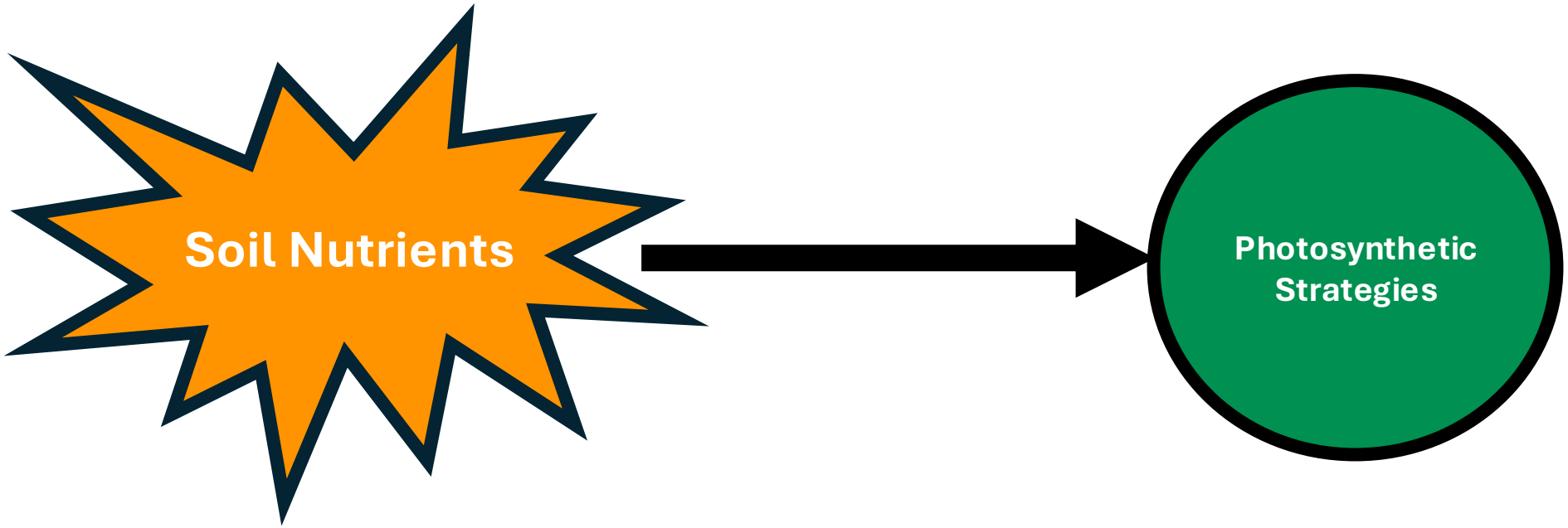
In cold/dry places, higher leaf nitrogen confers greater species population growth



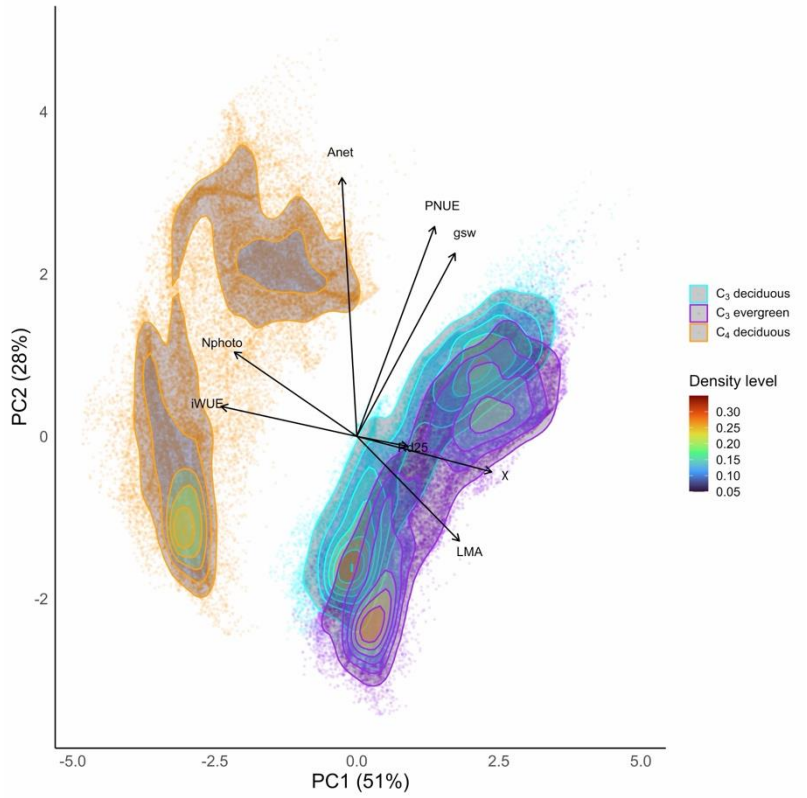
Cheaib…and Smith (in Prep)

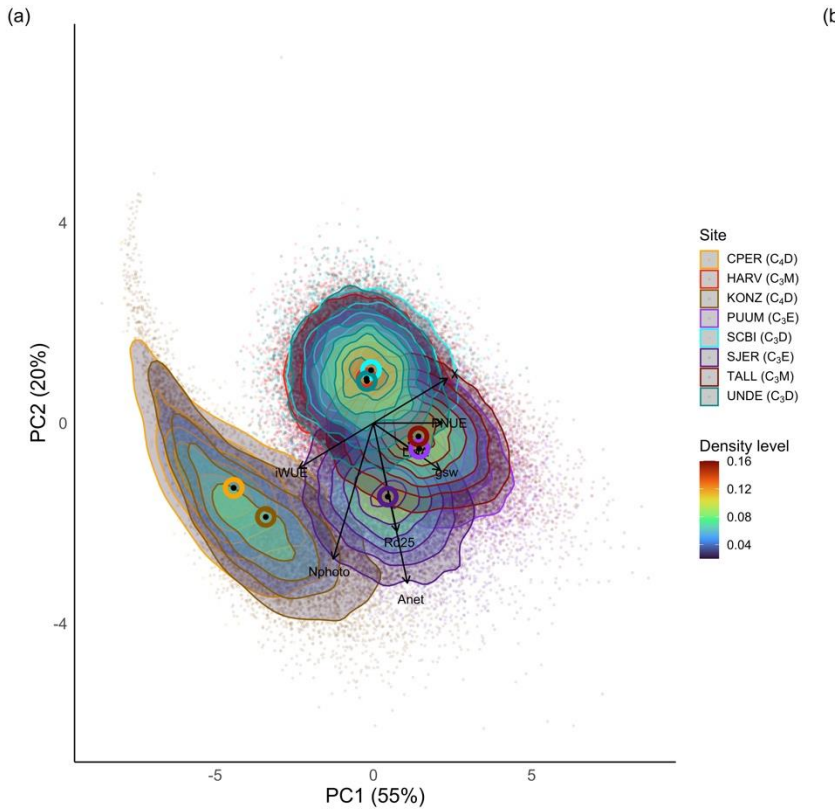
Perturbation vignettes

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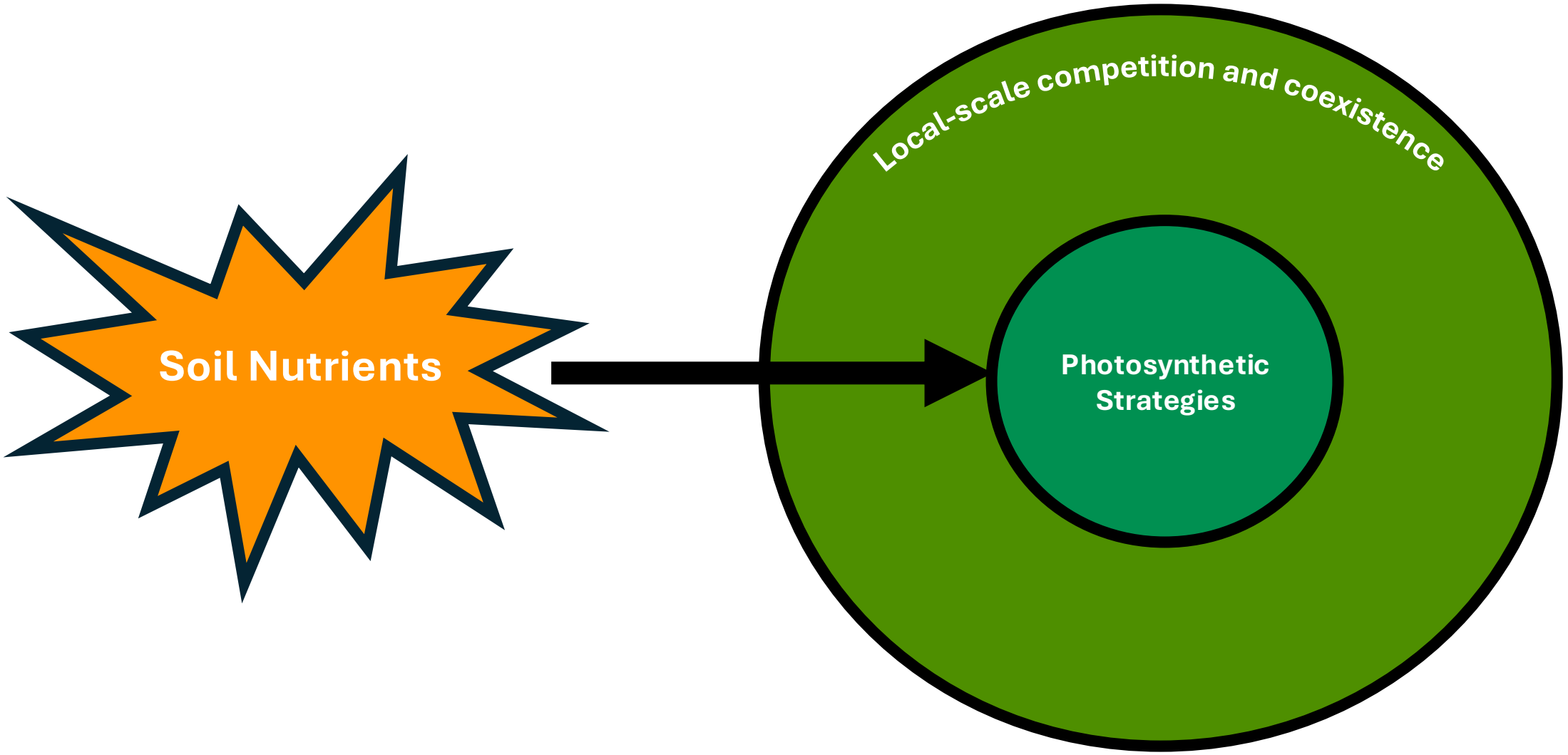

Soil Nutrients
Photosynthetic Strategies

We can predict global and local photosynthetic strategies from theory

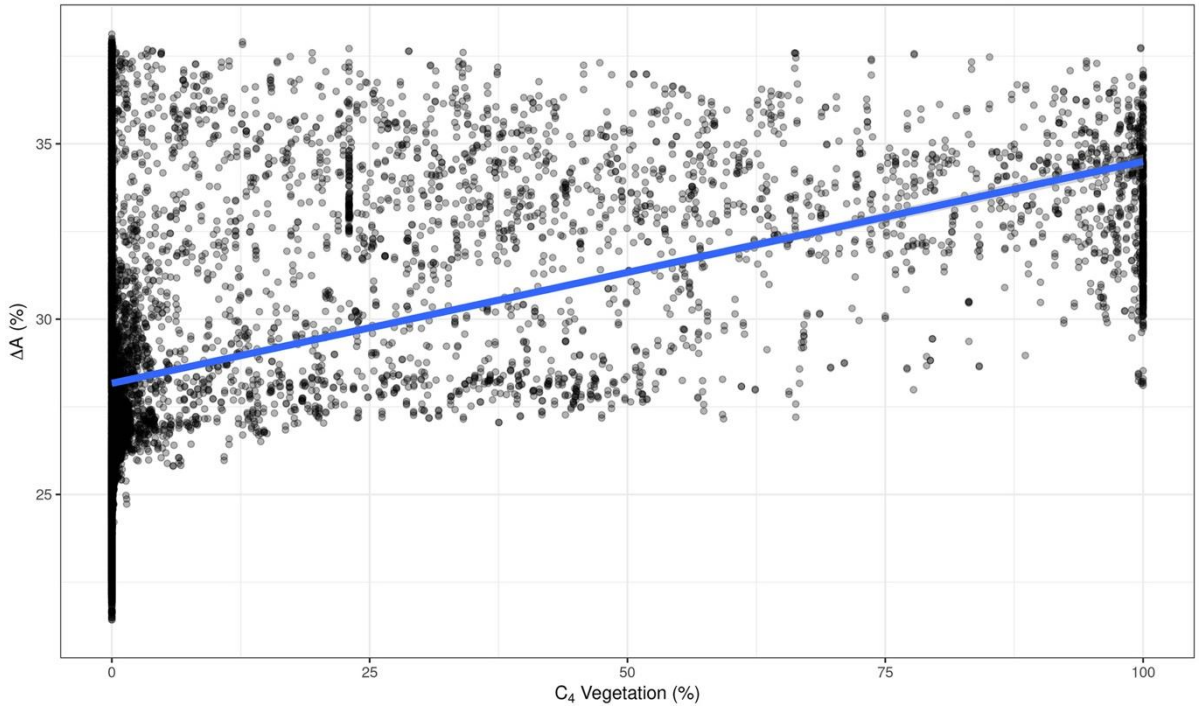


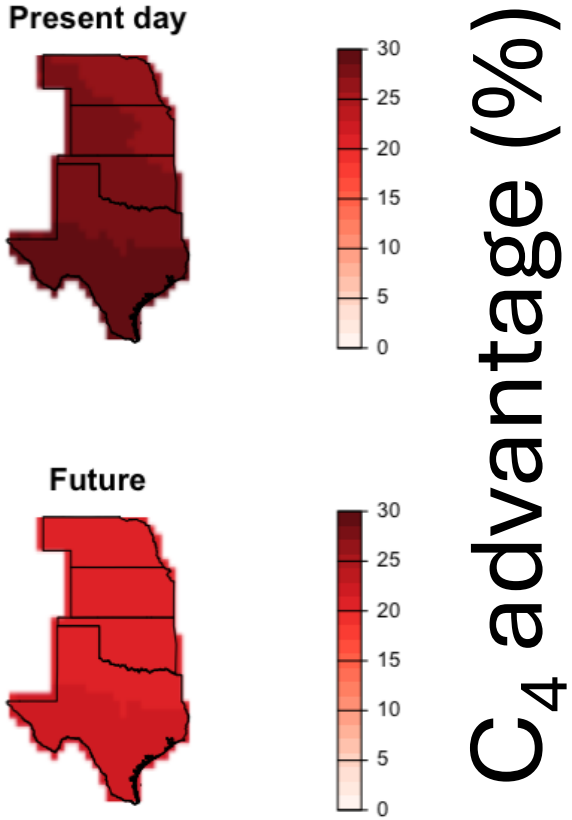


Smith (In Review)


Soil Nutrients
Local-scale competition and coexistence
Photosynthetic Strategies

Strategies can predict observable outcomes of competition (broadly)







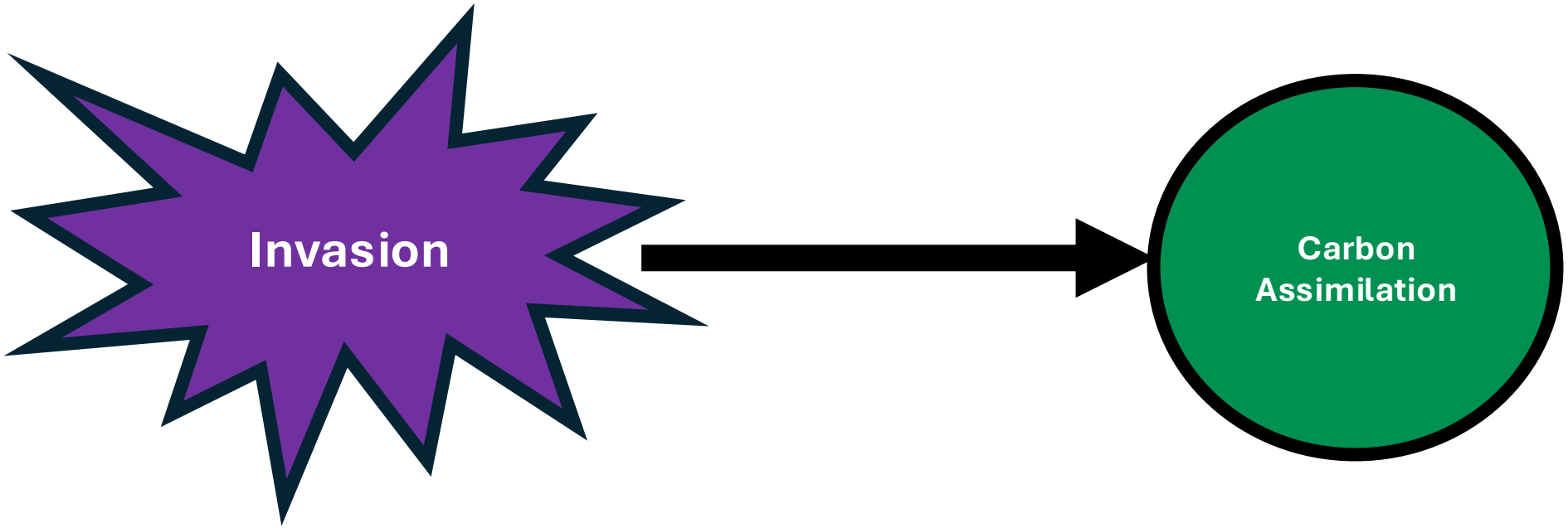




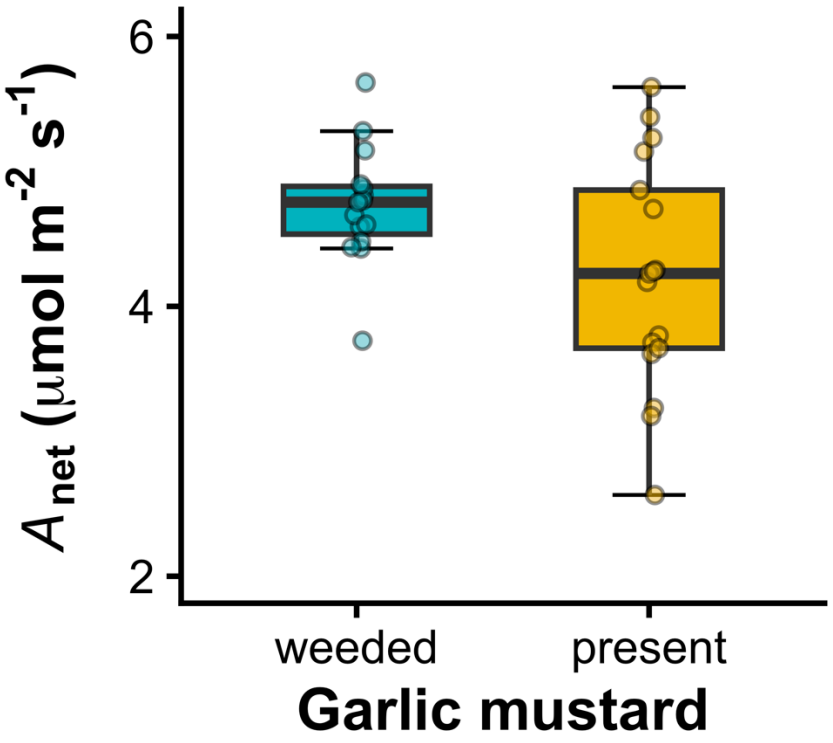
Scott and Smith (2022) *JAMES*

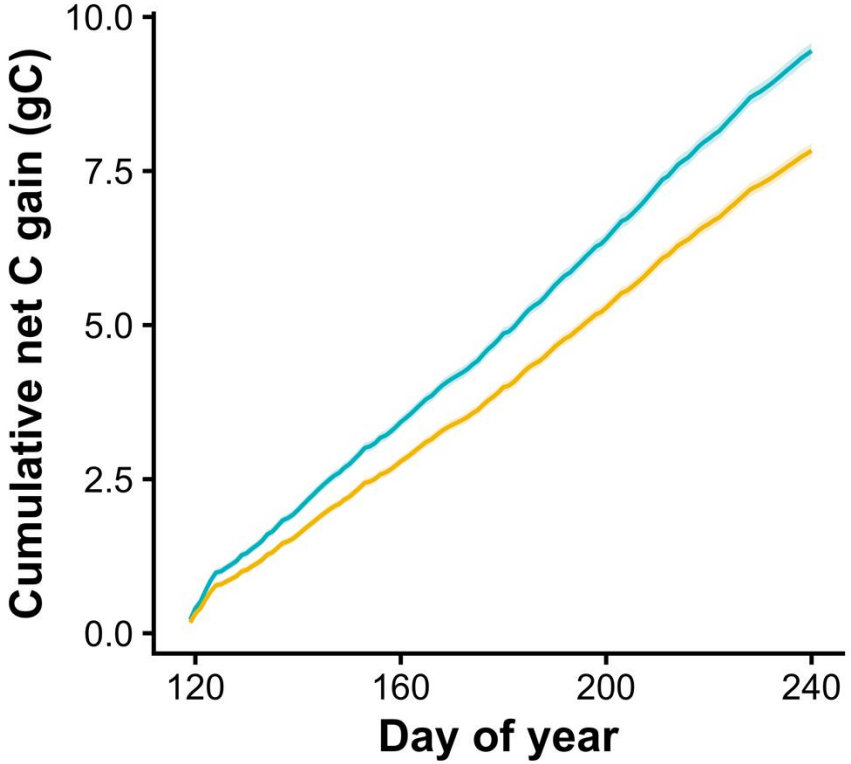
Perturbation vignettes

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Invasion
Carbon
Assimilation

Invasion by garlic mustard reduces native understory plant carbon assimilation

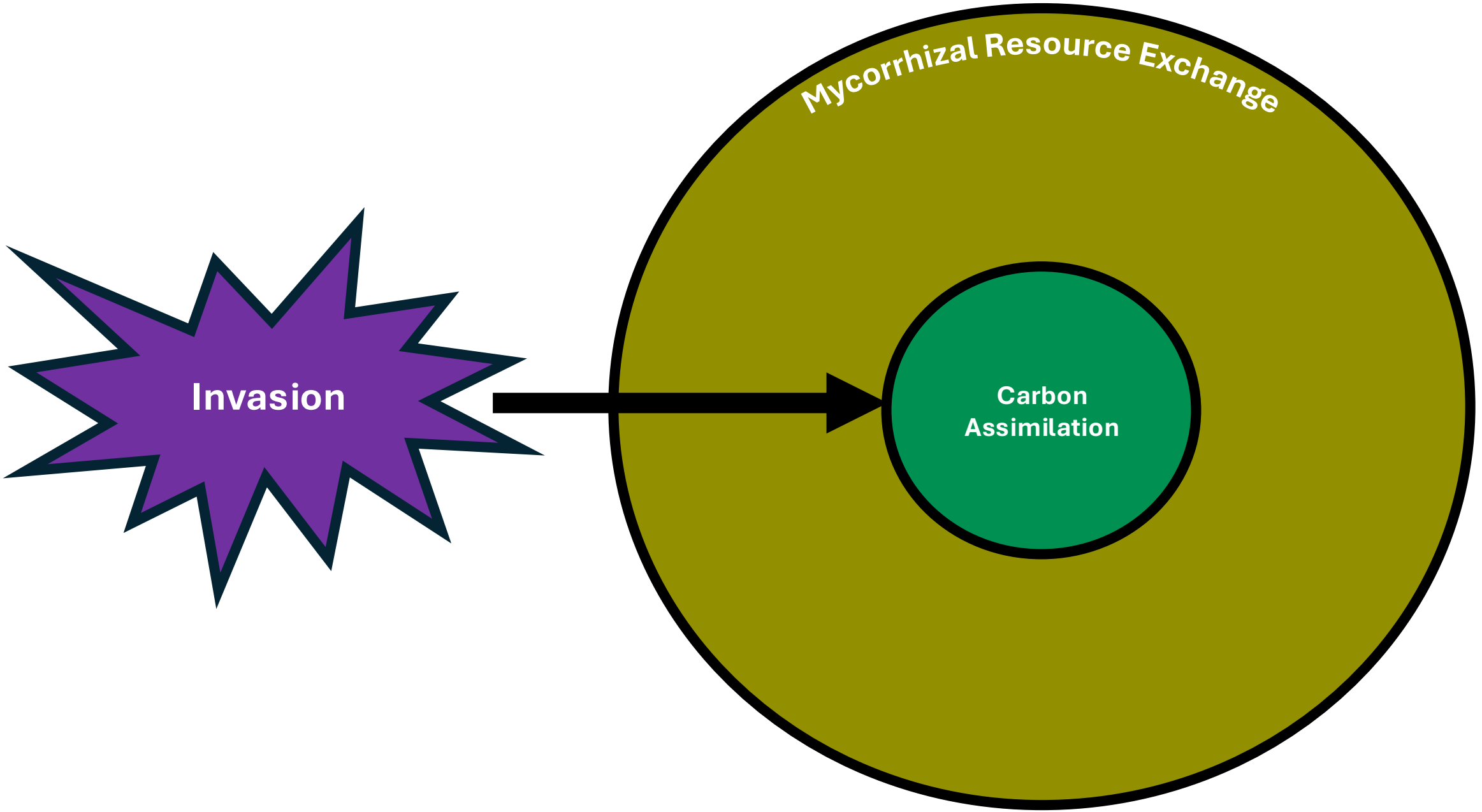






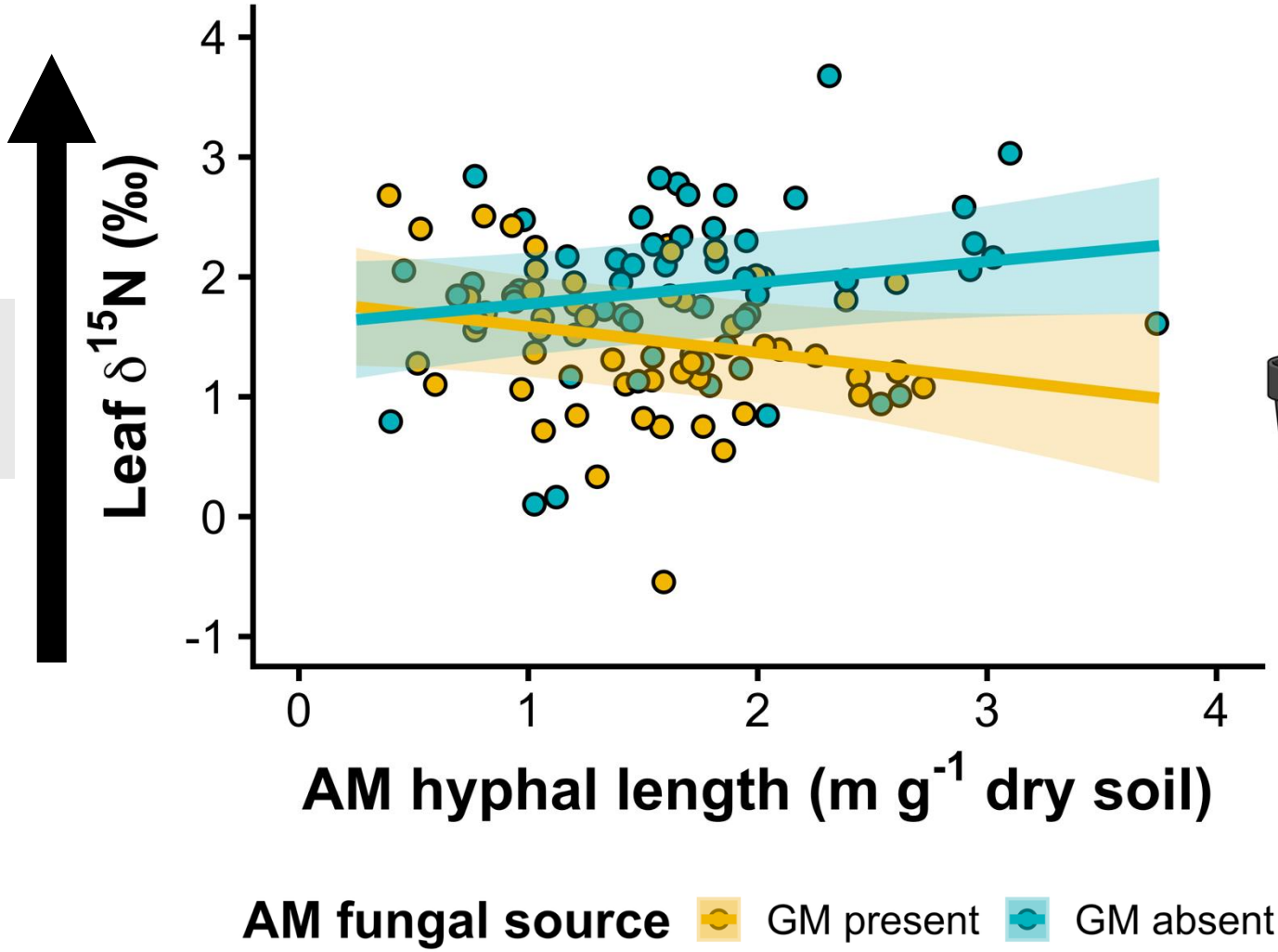


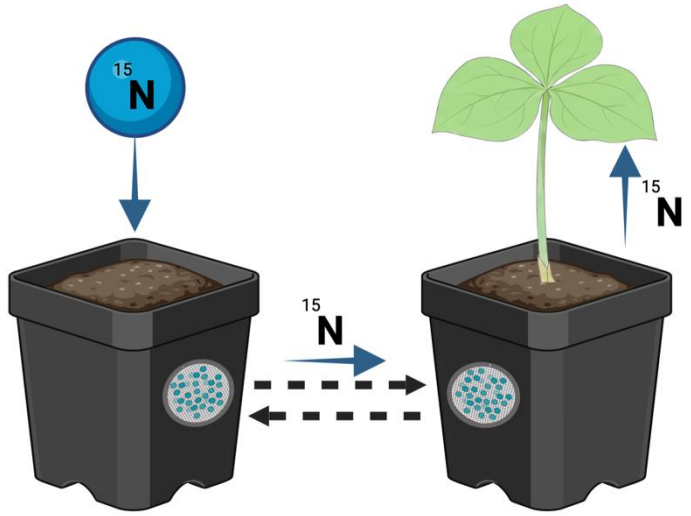
Perkowski…and Smith (2026) *Func Ecol*


Mycorrhizal Resource Exchange
Invasion
Carbon
Assimilation

Invasion impacts are due to mycorrhizal mutualism disruption

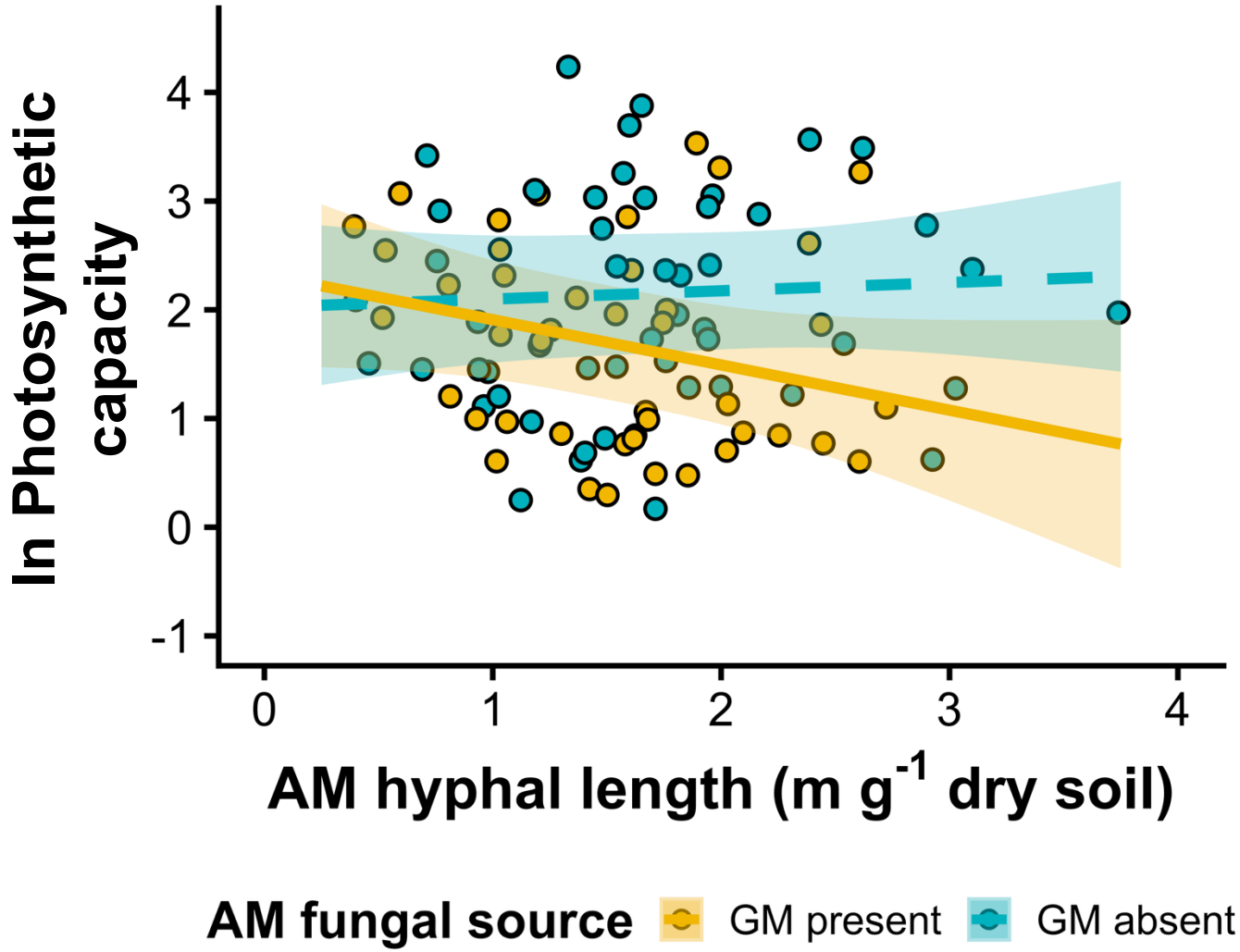
Greater share of leaf N from AM fungi





Perkowski…and Smith (In Prep)

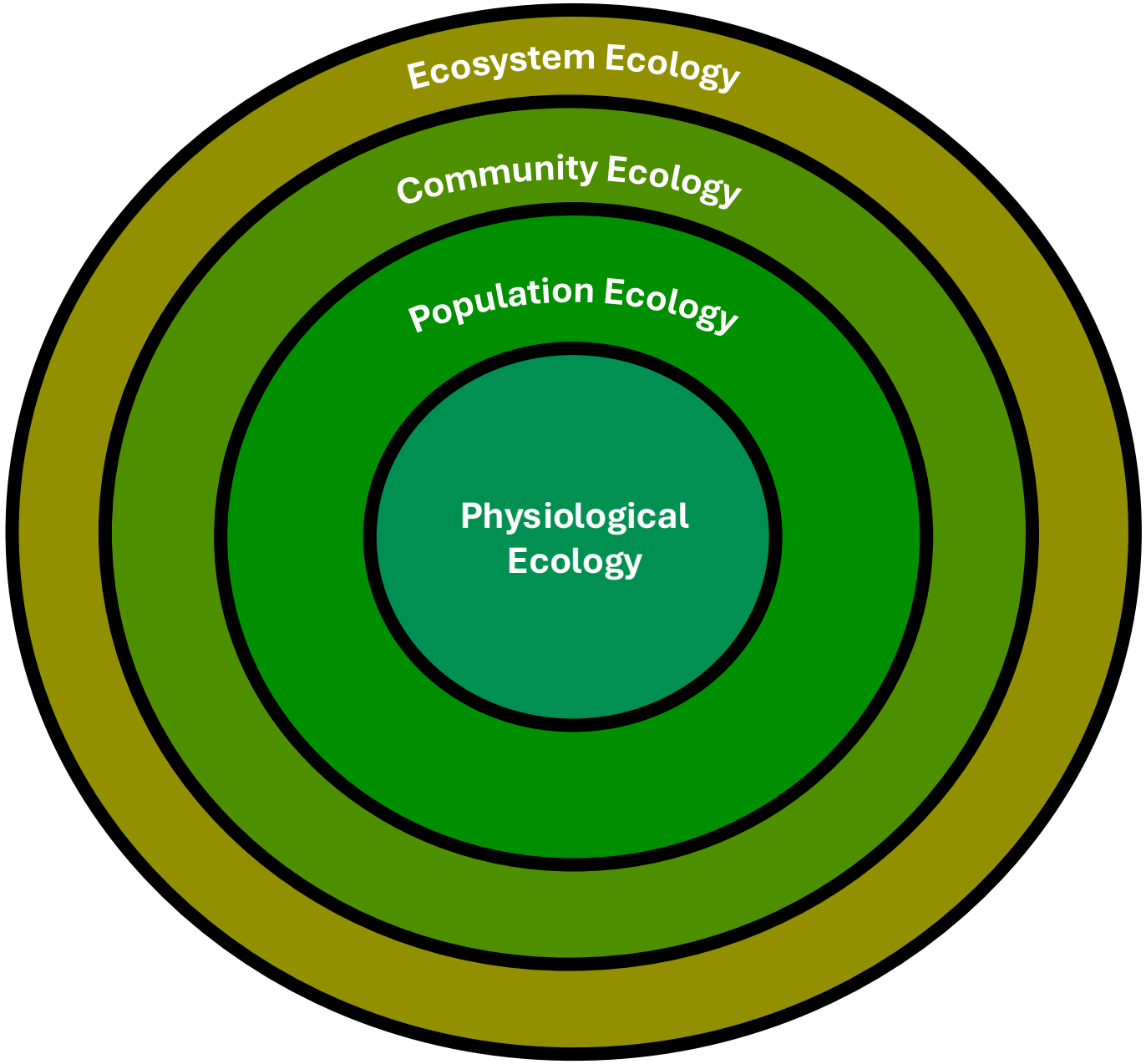
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Perkowski…and Smith (In Prep)

Conclusions

- Plant physiological responses to their environment form the basis of larger-scale responses
- Experiments, observations, theory, and modeling are needed to understand and predict these responses across scales








Presentation available at: github.com/SmithEcophysLab/seminar/
More about the lab: smithecophyslab.com
Contact: nsmith87@unl.edu

